UNIT 6 Internet and WWW

Internet is a network of networks. Millions of computer all over the world are connected through the internet. Computer users on the internet can contact with one another anywhere in the world. If a computer is connected to the internet, one can connect to millions of computers. It is very much similar to the telephone connection where one can talk with any person anywhere in the world.

On the internet, a huge resource of information is accessible to people across the world. Information in every field starting from education, Science, Health, Medicine, History and Geography to business, news, etc. can be retrieved from the internet. One can also download programs and software packages from the internet.

Due to the tremendous information resources the internet can provide, it is now indispensable to every organization and personal activities.

Some of the popular services of the internet are:

1. E-mail: Sending and receiving mail electronically.
2. File Transfer: Transferring files from one computer to another.
3. WWW (World Wide Web): Retrieving information residing on the internet servers in the form of websites.
4. Chat: Exchanging views or communicating information instantly.

The development of internet started when US Defense Department set up the ARPANET (Advanced Research Project Agency Network) to have a failure proof communication network for defense department of US.

This architecture was later adopted by an educational institute for an exchange of views among research scholars and then, it was thrown open to the public. Since 1994, the internet has grown by leaps and bounds driven by cheaper cost, easier to use and increase in information.

What are the Applications of Internet?

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1) On-line communication:

Computer users around the world use the E-mail services to communicate with each other extensively.

2) Feedback about products:

Commercial organizations are also using the internet to gather information about the satisfaction of existing products and market opportunities of new products.

This is usually accomplished by putting up an interactive survey application by the organization on a WWW site on the Internet.

3) Product promotion:

Several commercial organizations are effectively using the internet services for promoting their products by the use of different social networks.

4) Customer Support Service:

Many organizations are also using the internet to provide timely customer support.

5) On-line shopping:

The Internet has also facilitated the introduction of a new market concept, which consists of virtual shops. These shops remain open 24 hrs all the year round and are accessible to make purchase all around the world.

6) On-line journals and magazines:

There are many WWW sites on the internet, which consists of an electronic version of many journals and magazines.

7) Real-time updates:

It helps to provide news and other happenings that may be on-going in different parts of the word but with the use of the internet, we come to know about the real-time updates in every field be it in business, sports, finance, politics, entertainment and others very easily.

Many time the decisions are taken on the real-time updates that are happening in the various parts of the world and for this, the internet is very essential and helpful.

8) Research:

In order to do research, we need to go through hundreds of books as well as the references and that was one of the most difficult jobs to do earlier.

Since, the internet came into life, everything is available in just a click. The user just has to search for the concerned topic and will get hundreds of references that may be beneficial for the research and since, the internet is here to make research activity easy and hence, public user can take a large amount benefit from the research work that have been done.

9) Education:

Education is one of the best things that the internet can provide. There are a number of books, reference books, online help centers, expert`s views and other study oriented material on the internet that can make the learning process very easier as well as a fun to learn.

10) Financial Transaction:

It is a term which is used when there is an exchange of money. With the use of internet in the financial transaction, the work has become a lot easier. Payments, Funds transfer, banking transactions can be done through on-line banking service.

11) Entertainment:

The Internet is also used for entertainment. Such as chatting with friends, sharing videos, watching movies, listening music, live telecast of sports and other events, playing games, etc.

12) Job Search:

Using internet, searching job has become an easier task. There are an endless amount of websites on the internet that provided news about a vacancy in various post as required.

13) Blogging:

There are many people who are very much interested in writing blogs and for them the internet is the best place. They can not only write blogs as per their wish but can also publicize their work so that their work reaches to most of the people and they get appreciated.

Positive and Negative Impact of Internet

The positive impacts of internet to the society, organization and individual are:

1. Faster, cheaper and easier medium of communication.
2. Information sharing and browsing.
3. File transferring facility.
4. Reach to the worldwide viewers.
5. Effective, easier, faster and cheaper promotion of product or service.
6. Better customer support and customer relationship management (CRM).
7. Online services like banking, shopping, education, etc.
8. E-mail communication for sending and receiving an electronic document.
9. Enhanced collaboration between different organizations.
10. Effective Supply Chain Management (SCM).
11. Electronic payment system using credit /debit cards, ATM, online payment, electronic cheque,
    smart card, electronic purse, etc.
12. Newsgroups for instant sharing of news and feedback system.
13. Creation of new job opportunities related with the internet.
14. Source for entertainment.
15. Social networking for instant touch with friends and relatives.

Misuses of Internet / Negative impacts of Internet

The negative impacts of internet to the society, organization and individual are:

1. It is the most common medium for spreading malicious software like virus, worm, etc.
2. It has increased piracy.
3. Pornography (uploading, publishing, viewing sexual contents in the form of text, image, audio
   and video).
4. Stealing, modifying or destructing data.
5. Piracy of software, audio, video or other intellectual contents.
6. Hacking of organizational system, website, database etc.

7. It is also used to harass people by sending insulting comments, making vulgar cartoons,
blackmailing, etc.
8. Unemployment problem for the individuals not having knowledge about the Internet.
9. It has increased digital divide.

Connecting to the internet

Before Connecting

1

Activate your internet. If you’ve just moved somewhere new, the internet might not be active. To set up your internet, call your internet service provider (ISP) to turn on the internet connection for your residence.

2

Ensure that the source of the internet is on. It may seem obvious, but a common mistake that's often made when connecting to the internet is not making sure the source of the internet is on. Especially if you've just set up a router and/or modem, ensure that it's on and that everything is plugged in properly, and that any lights on it aren't indicating that there are problems.

- Cords can also be unplugged or slightly pulled out of the wall, rendering the operation futile. Make sure that everything is plugged in properly and is working right before getting started.

3

Understand that most mobile devices can only connect to wireless broadband. Devices like smartphones, tablets, iPods, handheld gaming systems, and so forth can usually only connect to Wi-Fi services, due to the portable nature of them. Therefore, you won't be able to connect a mobile device to ethernet or to a dial-up network. Ethernet and dial-up connections are limited to non-portable gaming devices (not covered in this article) and computers.

4

Know what "path" to take to get to your network settings. Regardless of what operating system or device you're using, you'll probably need to access your network settings at some point in the process. The process is slightly different for every device, but the general path that you'll need to take to access your network settings is usually the same, depending on the OS. Some common devices or operating systems, and their paths to the settings, are listed below.

- *Windows XP:* Start –> Control Panel –> Network and Internet Connections
- *Windows Vista:* Start –> Network –> Network and Sharing Center
- *Windows 7:* Start –> Control Panel –> Network and Internet
- *Windows 8:* Start –> Search "View network connections" –> View Network Connections
- *Windows 10/11:* Search "View network connections" –> View Network Connections
- *Mac OS X Jaguar and later:* System Preferences –> Network
- *Ubuntu and Fedora:* Network Manager
- *iOS (iPhone, iPad, etc.):* Settings –> Wi-Fi
- *Android:* Settings –> Wi-Fi (or Wireless & Networks)
- *Windows phone:* Settings –> Wi-Fi

Connect with Wireless Broadband

1. 1

Find your device's network settings. The Wi-Fi options can be found in the settings menu of mobile devices and the taskbar on computers. This is where you'll turn on the Wi-Fi and connect to a network. Here are a few common network settings locations:

- Windows — Click the Wi-Fi icon in the taskbar to switch on Wi-Fi.
- macOS — Click the Wi-Fi icon in the menu bar to switch on Wi-Fi.
- Android — Go to **Settings > Network & internet > Internet** to turn on Wi-Fi.
- iOS — Go to **Settings > Wi-Fi** to turn on Wi-Fi.

2

Make sure that the Wi-Fi connection for your device is on. Regardless of the device, it's possible to turn off Wi-Fi. Some devices have a physical switch that turns on or turns off the Wi-Fi, while others just have the ability to toggle Wi-Fi on the software settings. Make sure that the computer does not have the Wi-Fi capability turned off before proceeding.

- For example, some laptop models have a physical Wi-Fi button in the function row at the top of the keyboard. This key usually has an icon with three curved lines radiating from a dot.

3

Go to the list of nearby Wi-Fi networks. The Wi-Fi menu on your device will have a list of nearby networks that you can connect to.

- On mobile, navigate to your device's settings menu, then navigate to the network settings.
- On desktop, click the Wi-Fi icon in the taskbar (Windows) or menu bar (Mac).

4

Find the name of your Wi-Fi network. Your broadband network's router should have the default name printed on it. The name of a hotspot network will usually show up by default as the name of your cellular device (e.g. "[Your name]'s iPhone"). Find this name and select it.

- Wi-Fi or hotspot names can be changed, but if you've changed the name of your network or hotspot, you probably know what it is. If you weren't the one to change it, or you don't know what the name is, ask the person in charge of the network.
- Router Wi-Fi network names are typically on a sticker on the bottom or back of the router.

5

Enter the password to the network or hotspot. Some networks are public, but most aren't. If the network you're trying to connect to has a password, you'll be prompted for that password before you can connect to the network. The default password will usually be listed on the router, but if you don't know the password, ask the person in charge of the network.

- Some protected public networks may have varying passwords per person. For example, a school may allow students to log on to the network with their student ID number, rather than a single set password.
- See our guide to adding a password to your Wi-Fi if you want to change the default password.

6

Wait for the computer to connect. It often takes a few seconds for a device to connect to a wireless source, but if the computer can't establish the connection to the router, it will time out the connection attempt. In this case, move closer to the source, or disconnect and then reconnect your computer to the Wi-Fi.

- The connection attempt will also fail if the password has been entered incorrectly.

7

Test out your internet connection. Once you've connected to the internet, open up a page in your web browser and wait for it to load. Since some pages can crash, you may want to load up a reliable website, such as https://www.google.com/, to ensure that the website isn't going to be down.

- If your Internet is slow on one device, try joining the network on a different device to determine if there's an issue with your router.[1]

8

Troubleshoot if your computer won't connect to the internet. For some people, the Wi-Fi will connect without issue. But sometimes, that's not the case! There are many reasons that a computer may not be able to connect to the wireless connection — most computers have built-in software that can diagnose the problem. A few common problems are listed below:

- Some older computers are unable to connect to the internet wirelessly. You may need an ethernet cable to get online.
- If the internet is slow or won't connect, you may be out of range of the router or hotspot. Try moving closer to the source.
- If the network isn't showing up, you may be out of range, or the network may be down. Try moving closer or rebooting your router. You can also try putting your router up on a higher shelf to improve the signal.[2]

Connect with Ethernet

1

Get an ethernet cable and any needed adapters. Many recent devices can connect directly to the router via an ethernet cable. However, some aren't built to do that. Laptops, for example, often don't have a port for using ethernet. For that reason, make sure you get any adapters you may need for the ethernet cable to ensure that you can use it.

- Ethernet cables are all different; for example, a Cat-5 or Cat-5e cable runs at slower speeds than a Cat-6. However, it's also largely dependent on the router's connection and how many people will be connecting to the network at once. Unless you're doing very, very intensive upload work, you're probably not going to need a Cat-6 cable if you're the only one on the network.
- You cannot connect a mobile device (e.g. a smartphone) to ethernet with an adapter.

2. 2

Connect one end of the ethernet cable to the broadband source. The broadband source will most likely be a router, but in some cases, it may be a modem. In either case, you'll need to plug in one end of the ethernet cable to the broadband source to ensure that the computer will connect.

- One ethernet port on the router will already be filled by the modem’s ethernet connection. Select one of the other free ethernet ports on the router to connect your device to the internet.

3

Connect the other end of the cable to the computer. Find the ethernet jack on your computer and plug it in. This jack will typically be located on the back of the computer, where the other components plug in.

- If your computer doesn't support ethernet, you'll need to make sure the computer is connected to the adapter, and then connect the cord via the adapter. For example, you might need an ethernet-to-USB-C adapter if your laptop only has USB-C ports.

4

Access your computer's settings. You'll need to ensure that the computer is set to recognize the ethernet, rather than wireless. Most likely, you'll have to turn off your wireless connection to ensure that the computer recognizes the ethernet connection instead.

5

Test out your internet connection. Open up a page in a web browser and see if it loads. Some web pages can take longer to load than others, and others crash sometimes, so you may want to try and load a reliable website (e.g. https://www.google.com/) to ensure that the connection is running.

- For more ethernet tips, check out how to connect two computers together with an ethernet cable and protect an outdoor ethernet cable.

6

Troubleshoot if you can't connect. Ethernet is more reliable than Wi-Fi, but that doesn't mean that things still can't go wrong. If you're having trouble with the ethernet, it can stem from many problems, but make sure that the basics (e.g. the router being connected) are established, and that your computer isn't being troublesome.

- Make sure there's no problem with the ethernet cable (which can range from "the cord wasn't plugged in all the way" to "the cable is faulty/broken and needs to be replaced").
- Check if the router is having trouble, and reboot it if so. Contact your ISP if resetting the router doesn't work, but the cord and computer's ethernet connection work fine.
- Rarely, your computer's ethernet card may turn out to be defective. If this is the case, contact the seller of your computer or the computer's manufacturer.

Connect a Computer with Dial-Up

1

Understand that dial-up internet is no longer widely supported and it will be very difficult to do certain activities on the internet with this type of connection. With dial-up internet, you may be only limited to browsing websites that are mostly text and/or images without many add ons and features. Because dial-up internet has fallen out of use in favor of broadband internet, it's not common to see instructions for connecting to dial-up internet anymore. If you want to do some very serious internet browsing, it will be best to find a Wi-Fi hotspot in a public location. However, dial-up is still commonplace in a few rural areas, which means that you may find yourself needing to connect to it.

2

Ensure that you can connect to dial-up. Dial-up internet requires the use of a phone line, and can only connect one person per phone at a time. If somebody else is already connected, and/or the phone line is being used to make a call, you will be unable to connect until the other person disconnects or hangs up. Additionally, most new computers do not have the components to connect to dial-up; you may have to purchase an external USB modem so your computer can connect.

3

Plug in the modem to the phone jack. Oftentimes, places with dial-up internet will have two phone lines - one for the phone, and one for the modem. However, if the modem isn't often used, it may be unplugged, or there may only be one phone line. Make sure that the phone cable is plugged into both the phone jack on the wall, and the plug on the modem.

4

Connect the modem to the computer. Using another phone line, insert one end of the second phone cable into the modem and the other end into the computer's modem jack (or the converter).

- Make sure that you don't accidentally plug the phone cable into the ethernet plug by mistake. The phone jack on the computer should be noted by a small phone next to it.

5

Access your computer's network settings. You'll need to manually set up the dial-up connection on the computer. From there, configure the modem settings. If this is your first time connecting to the dial-up source, you'll most likely need to configure the modem's network settings. While the process is slightly different for every OS, you'll need to enter the same information: the dial-up phone number, a username, and a password. The settings paths that you'll need to follow in order to configure the network are:

- *On Windows XP:* Network and Internet Connections –> Set up or change your Internet connection –> Setup

- *On Windows Vista:* Network and Sharing Center –> Set up a connection or network –> Set up a dial-up connection
- *On Windows 7 and 8:* Network and Internet –> Network and Sharing Center –> Set up a new connection or network –> Connect to the Internet –> Dial-up
- *On Windows 10/11:* Network –> Dial-up Connection
- *On Mac OS X:* Network –> Internal/External Modem –> Configuration
- *On Ubuntu or Fedora:* Network Manager –> Connections –> Modem Connections –> Properties

6

Connect your computer's connection to the modem. If the dial-up settings are already configured, it may just be as simple as opening up the network settings and connecting to the modem, rather than searching for wireless connections. You will have to enter the number, username, and password, however.

7

Test out your internet connection. To ensure that your internet connection is working, open up a webpage and wait for it to load. Dial-up internet is much slower than typical broadband speeds, so don't be surprised if it takes some time. You may want to try and load a solely text-based webpage to increase the loading speed and tell if your internet is working.

- https://www.google.com/ is a reliable, simple website to try loading.

8

Troubleshoot if you can't connect. While dial-up is no longer widely supported, it's still possible to have issues with it. Ensure that the phone line is properly plugged in and that your system can connect to dial-up internet.

- Windows 10 has been known to have some trouble with dial-up connections sometimes. You may have to use a computer with an older operating system, if available.
- Ensure that you haven't accidentally plugged the phone cable into the ethernet jack by mistake. The phone cable's jack is smaller and is often denoted by a phone symbol

Client Server Technology

Client-server is a computer model that separates client and server, and usually interlinked using a computer network. Each instance of a client can send data requests to one of the servers online and expect a response.

In turn, some of the available servers can accept these requests, process them and return the result to the client. Although the concept be applied to various uses and applications, the architecture is almost the same.

Often clients and servers communicate through a computer network with separate hardware, but the client and server can reside on the same system. The machine is a host server that is running one or more server programs that share their resources with clients.

A client does not share its resources, but requests content from a server or service function. Clients, therefore, initiate communication sessions with the servers that wait for incoming requests.

Client-server – Description

The character of client-server describes the relationship of programs in an application. The server component provides a function or service to one or many clents, who start their service requests.

Functions such as exchanging email, Internet access and database access, are built based on client-server model. For example, a web browser is a client program running on a user’s computer that can access information stored on a web server on the

Internet. Users accessing banking services from your computer using a Web browser client to send a request to a web server in a bank.

That program may in turn forward the request to its own program database client that sends a request to a server database on another computer to retrieve bank account information. The balance is returned to the client database of the bank, which in turn serves it back to the client browser and displays the results to the user.

The client-server model has become one of the central ideas of network computing. Many business applications being written today use the client-server model. In marketing, the term has been used to distinguish distributed computing by smaller dispersed computers from the “computing” monolithic centralized mainframe computers.

Each instance of client software can send data requests to one or more servers connected. In turn, the servers can accept these requests, process them and return the requested information to the client. Although this concept can be applied to a variety of reasons for different types of applications, the architecture remains fundamentally the same.

Features of the Client

- Always start applications servers;
- Waits for responses;
- Get answers;
- Usually connects to a small number of servers at once;
- Normally, interacts directly with end users through any user interface, such as graphical user interface.

Server Features

- Always wait for a request from a client;
- Serves clients’ requests, then responds with the requested data to clients;

- A server can communicate with other servers in order to meet a client's request.

Client-server Architecture – Benefits

- In most cases, the client-server architecture enables the roles and responsibilities of a computing system to be distributed among several independent computers that are known to itself through a network. This creates an additional advantage to this architecture: greater ease of maintenance. For example, you can replace, repair, upgrade or even relocate a server clients, while continuing to be the conscience and not affected by this change;

- All data is stored on servers, which typically have far greater security controls than most clients. Servers can better control access and resources to ensure that only clients with appropriate permissions can access and change data;

- Since data storage is centralized, updates the data are much easier to administer, compared to the P2P paradigm, where a P2P architecture, data updates may need to be distributed and applied to each point in the network, which is the is time-consuming error-prone, as there may be thousands or even millions of peers;

- Many advanced technologies for client-server are now available that are designed to ensure safety, ease of user interface and ease of use;

- Works with many different clients of different capabilities.

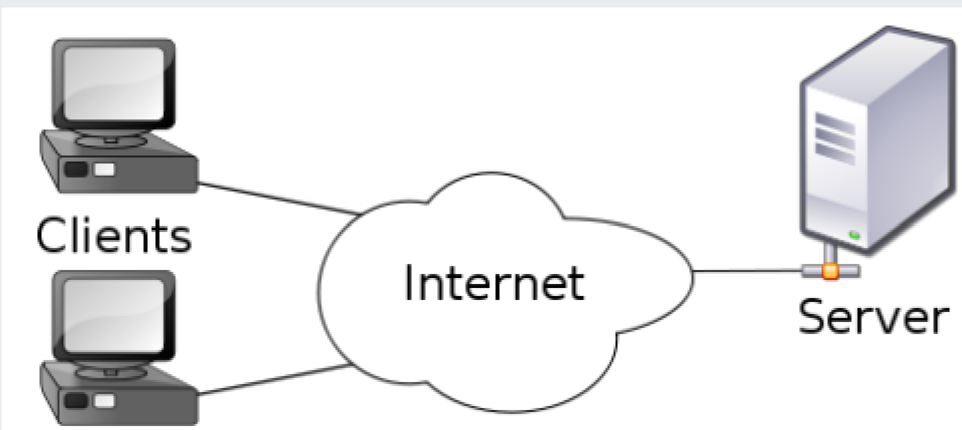
Client-server Architecture – Disadvantages

- Network traffic blocking is one of the problems related to client-server model. As the number of simultaneous requests the client to a particular server, the server may become overloaded;

- The client-server paradigm lacks the robustness of a P2P network. Under client-server, if a critical server fails, the clients' requests can not be met. In P2P networks, resources are usually distributed among multiple nodes. Even if one or more nodes depart and abandon a downloading file, for example, the remaining nodes should still have the data necessary to complete the download

Internet as a client server technology

Client-server is a computer model that separates client and server, and usually interlinked using a computer network. Each instance of a client can send data requests to one of the servers online and expect a response. In turn, some of the available servers can accept these requests, process them and return the result to the client. Often clients and servers communicate through a computer network with separate hardware, but the client and server can reside on the same system. The machine is a host server that is running one or more server programs that share their resources with clients. A client does not share its resources, but requests content from a server or service function. Clients, therefore, initiate communication sessions with the servers that wait for incoming requests.



Internet is a client/server technology because most services that are *accessed via* the Internet work on the client/server model. In that, we have a server which provides a service, while clients connect to the server to make requests regarding the service. A web site is an example: a web server holds the contents of the web site. Our web browser is a client, which we can use to connect to web servers to ask them for information, or to perform services on our behalf.

Electronic Mail (e-mail) is one of most widely used services of Internet. This service allows an Internet user to send a **message in formatted manner (mail)** to other Internet user in any part of the world. Message in mail not only contain text, but it also contains images, audio and videos data. The person who is sending mail is called **sender** and person who receives mail is called **recipient**. It is just like postal mail service.

Advantages of E-mail :

1. E-mails provides faster and easy mean of communication. One can send message to any person at any place of world by just clicking mouse.
2. Various folders and sub-folders can be created within inbox of mail, so it provide management of messages.
3. It is effective and cheap means of communication because single message can be send to multiple people at same time.
4. E-mails are very easy to filter. User according to his/her priority can prioritize e-mail by specifying subject of e-mail.
5. E-mail is not just only for textual message. One can send any kind of multimedia within mail.
6. E-mail can be send at any hour of day, thus ensures timeliness of message.
7. It is secure and reliable method to deliver our message.
8. It also provide facility for edition and formatting of textual messages.
9. There is also facility of auto-responders in e-mail i.e. to send automated e-mails with certain text.
10. To write an e-mail there is no need of any kind of paper, thus it is environment friendly.

Disadvantages of E-mail :

1. It is source of viruses. It is capable to harm one's computer and read out user's e-mail address book and send themselves to number of people around the world.
2. It can be source of various spams. These spam mails can fill up inbox and to deletion of these mail consumes lot of time.
3. It is informal method of communication. The documents those require signatures are not managed by e-mail.
4. To use facility of e-mail, user must have an access to internet and there are many parts of world where people does not have access to Internet.
5. Sometimes, e-mails becomes misunderstood as it is not capable of expressing emotions.
6. To be updated, user have to check inbox from time-to-time.

***Video conferencing** solutions provide a live, visual connection between two or more people who are usually located in separate locations. The purpose of live video conferencing services is essential to emulate face-to-face conversations over the internet.*

Advantages

As you can image, there are a lot of advantages for corporates when they use the services of live video conferencing companies. Let's take a look at some of the significant and prominent benefits of using video conferencing tools for communication and collaboration.

1. Saves time and resources –

For any businesses to run properly, it is important for efficient and effective communication to exists in the organization. In the earlier days, the only way different offices could communicate effectively was by having meetings together, which required the employees to travel a lot.

While conference calls were a possibility even in those days, it is not as effective as live video conferencing services these days. By providing a cheaper and more effective way of communication, video conferencing tools save a lot of time and resources for the company.

2. Increases productivity of the employees –

Live video conferencing companies and video conferencing service providers have made internal communication easier, faster, and more convenient than ever. Due to video conferencing tools, anyone or any team in the entire organization can communicate with any other team or employee. This instantaneous and effortless communication has effects that carry over into the productivity and performance of the employee.

By using the services of video conferencing companies, most businesses are not only improving their overall internal communication but are also increasing the productivity of each employee in the organization.

3. It has a lot of intangible benefits too –

Along with increasing the productivity of the organization in general, live conferencing tools also have a lot of intangible benefits for any organization such as a strong sense of community due to the personal face-to-face conversation instead of only hearing the voice of the person they're communicating with.

4. Video Conferencing Improves Communication

No matter what the size of the organization is. Clear concise and effective communication is what the requirement of every project is so that it can be completed properly and delivered timely. A Study suggests that visual memories are the strongest memories, and samely visuals are faster to process compared to any audio or text, as it takes a little sense of personalisation in it. Video conferencing effectively initiates quality communication and enhances the degree of understanding.

5. Helps build relationships

Just like in-person meetings, video conferencing also gives you the chance to build and foster relationships. As emails, chats, and audio calls can't make it possible to build that trust and level of connection live video conferencing did. The technology of video communication has replaced the need to travel from one to another city for meetings with your clients or to coordinate with your employees and teams. As it helps bridge the gap, there are many times you may need to travel for holidays or for some important purposes, still from any city you can attend your important meetings. When you are always available for someone to help and that too in this effective way, it is easier for you to build relationships with people and that brings harmony to the world.

6. Bring Collaborations

Video conferencing comes up with lots of features that increase the overall interactivity of connected people. How? As said the features are so many like you can be able to share your screen, can edit the document in real-time and whatnot. Additionally, you can video conference with your other partners or bring them into the team discussion like an expert panel and can examine or share any type of document. Nowadays, everyone is not working from the office only as there are new types like hybrid, part-time, online, freelance and other modes have been developed. Making it possible for a collaborative meeting and discussion from their respective locations.

7. Improves Productivity

Video conferencing is more effective. As it saves everyone's time because of the clearer communication provided by vocal and nonverbal cues, screen sharing, real-time collaboration, and the ability to join from any geographical location. Video conferencing can be scheduled at a fixed time and being on time increases productivity while saving lots of time and also the ability to work flexibly with this increases willingness. Additionally, it is now easy to clear any query of your client and a reduced doubt client can be the best one to potentially purchase making it possible to increase your business ROI.

8. Makes Meeting Easier

Scheduling meetings can be challenging if there is a lot of travel involved within your team. But the very own nature of video communication makes it possible to connect with the team and join the meeting practically from anywhere. Regardless of the location they just need a system or gadgets like a laptop or a smartphone with an internet connection. These two configurations are enough to make them connected from the airport, hotel room, taxi or other places. Nowadays, there are features like integrated calendars and workflow sheets for the team.

9. Manageable Records

Face-to-face meetings can be sometimes hard as well as making notes with an audio call is quite difficult. While with video conferencing you can record it and share it with your team, so they can refer to that video for any doubts or queries effectively. Additionally, if there is someone who was absent from the meeting can get the insights and help you avoid multiple scheduling of meetings leading to saving time and effort.

10. One-Stop Solution

Video conferencing is quite an effective one-stop solution for those who want more than just having meetings with the platform. Simply referring to the effectiveness and the feature-loaded video communication technology. How? As the tools of this technology make it possible to enable live events like webinar panel discussions or even product launch events too. The ability to share video messages and thoughts virtually leads to a greater capability of targeting the audience of the world as well as your team and employees.

Disadvantages

As it the case with any technology, video conferencing also doesn't come without a few caveats and disadvantages. Even though most people would correctly argue that when it comes to video conferencing solutions, the advantages weigh out the disadvantages but in order to have a well-developed outlook on anything, it is important to see both sides of the argument.

Every technology has pros and cons, so why are we here with the disadvantage of video conferencing? Make you aware of the very fact whether this video communication technology will be helpful for you or not.

Here are a few disadvantages that are typically faced by corporates when it comes to using the services of live video conferencing service providers.

1. It still lacks the personal touch of face-to-face communication –

While most conferencing solutions by live video conferencing vendors come really close to emulating the experience of personal face-to-face communication online, there's still a small gap between the effectiveness of video conferencing and a face-to-face meeting.

Most corporates understand this and usually also have face-to-face meetings between teams on important issues. However, whereas these meetings used to be the primary method of inter-organization communication, video conferencing service providers have definitely reduced the frequency of these meetings in general.

2. Even the best systems can suffer from technical problems –

No technology is impenetrable to glitches and problems. While most live video conferencing vendors provide 24×7 support for any corporate system they install, glitches can still happen, which wastes a lot of time and resources for the company.

3. It has a high initial cost –

Setting up an enterprise-grade video conferencing solution is often an expensive task as most solutions require specialized hardware that also needs to be set up, often at multiple locations. Considering the quality of gear and the expertise needed to set up the entire system, it is natural that the process has a high initial cost. However, most corporates are more than willing to pay the costs as it is recovered within a pretty short time of operating.

The disadvantages and advantages of video conferencing are both parts of technology but how effectively we utilize it in making a significant impact in all our lives is a crucial task. If you are a big enterprise that is looking for a solution which can improve the productivity of your team and make it possible to connect in a seamless way, then this is a great chance for you to adopt the best video conferencing technology.

What Is An ISP?

To simply answer "What is an ISP?", it refers to a company that provides internet connection to both residential and commercial users. ISPs enable their clients to use the internet for various purposes, like browsing, facilitating business operations and shopping and communicating with friends and family for a subscription cost. ISPs may also offer email services, domain registration and web hosting, among other services. Depending on the services it provides, an ISP can also be an information service provider, a storage service provider, an internet service provider or any combination of the three.

How An ISP Works

Multiple ISPs may link with many high-speed internet connections. Larger ISPs have their own high-speed leased lines, reducing their dependence on telecommunications providers and allowing them to deliver superior customer service. ISPs further maintain thousands of servers in data centres. The number of servers varies based on the ISP's coverage area. These enormous data centres oversee all consumer traffic. Following are three major categories of ISPs:

- **Tier 1 ISPs:** These ISPs have the largest global reach and own sufficient network lines to transmit most traffic independently. They do not pay anyone for the internet but invest heavily in setting up the infrastructure.
- **Tier 2 ISPs:** These ISPs have a country or region-wide reach and connect tier 1 and tier 3 ISPs as service providers. They pay for internet access to tier 1 networks, but also partner with other tier 2 ISPs for bandwidth exchange.
- **Tier 3 ISPs:** These ISPs link end-users to the internet via the network of another ISP they rely on and pay higher-tier ISPs for internet service access. They are primarily concerned with providing Internet connections to local businesses and consumer markets.

Classification Of ISP

Following are the distinct categories of internet service providers:

- **Access provider**: Internet access providers pay subscription fees to end service providers and give the end-user access to internet connections, including dial-up, broadband cable wi-fi, fibre optics, subscriber lines, cable modems, integrated services, digital networks and satellites.
- **Transit ISP:** Transit ISPs offer a large bandwidth to ensure hosting and access ISPs to maintain internet service. These providers pay for the transmission to the ISP above them and charge their clients for the internet as they use it.
- **Mailbox provider**: Mailbox providers ISPs offer hosting domain mailboxes for receiving and storing emails.

- **Hosting ISP:** Hosting ISPs offer a range of facilities and services, including email, cloud, web hosting, virtual servers and physical server operation services.
- **Free ISP:** Free ISPs offer their clients a variety of unique telecom services for free to access the internet. These free ISP businesses often show adverts to their internet users.
- **Virtual ISP:** They purchase services from another ISP. Customers of virtual ISP connect to internet service under their brand, while the wholesale ISP operate and maintain the infrastructure.
- **Wireless ISP:** Internet service providers (ISPs) that use wireless technology offer internet access using a wireless network system. ISPs enable their clients to connect to their servers via wireless connections, such as wi-fi enabled cellphones and laptops, at the convenience of tethering hotspots.

Types Of ISP Connection

A service provider may offer multiple types of internet service, often known as connection types. The most common types of internet connection include:

Fibre

This service employs glass-based fibre optic cable to transport data at the speed of light. As a result, the fibre connection type offers one of the fastest download and upload speeds. It also offers the lowest latency of any internet protocol, resulting in fewer delays for videoconferencing and online gaming. Fibre enables intensive internet use. Multiple users can concurrently stream video, play live-action games and share large files, along with connecting a variety of personal and household gadgets.

DSL

A digital subscriber line (DSL) establishes a connection to the Internet over a telephone line. As most houses already have a telephone connection, DSL is easily available and often offered by existing telephone companies. It is important to consider that with increasing distance between the user and the ISP's facilities, the internet speed may decrease because of a rise in latency. This means the connection becomes weaker as the signal interference grows with distance.

Cable

Coaxial cable helps provide the internet for this type of connection. It is the same wire that transmits cable television to a residence and most cable TV companies also offer cable internet access. Cable internet is often fast and dependable. It also has low latency, so users usually have fewer delays and ping loss, such as when playing online games.

Satellite

This wireless internet service transmits and receives data between your home and the internet via geostationary satellites. With this type of connection, the data travels to and from space. It has the greatest latency or delay of all connection types. The weather and network traffic can also impact its internet speed. Sometimes, there is a specified data cap that limits the total data you may use. The combination of these disadvantages usually makes satellite internet unsuitable for real-time gaming and video streaming.

What is DNS?

The Domain Name System (DNS) is the phonebook of the Internet. Humans access information online through domain names, like nytimes.com or espn.com. Web browsers interact through Internet Protocol (IP) addresses. DNS translates domain names to IP addresses so browsers can load Internet resources.

Each device connected to the Internet has a unique IP address which other machines use to find the device. DNS servers eliminate the need for humans to memorize IP addresses such as 192.168.1.1 (in IPv4), or more complex newer alphanumeric IP addresses such as 2400:cb00:2048:1::c629:d7a2 (in IPv6).

How does DNS work?

The process of DNS resolution involves converting a hostname (such as www.example.com) into a computer-friendly IP address (such as 192.168.1.1). An IP address is given to each device on the Internet, and that address is necessary to find the appropriate Internet device - like a street address is used to find a particular home. When a user wants to load a webpage, a translation must occur between what a user types into their web browser (example.com) and the machine-friendly address necessary to locate the example.com webpage.

There are 4 DNS servers involved in loading a webpage: (Types of DNS Servers)

- DNS recursor - The recursor can be thought of as a librarian who is asked to go find a particular book somewhere in a library. The DNS recursor is a server designed to receive queries from client machines through applications such as web browsers. Typically the recursor is then responsible for making additional requests in order to satisfy the client's DNS query.

- **Root nameserver** - The root server is the first step in translating (resolving) human readable host names into IP addresses. It can be thought of like an index in a library that points to different racks of books - typically it serves as a reference to other more specific locations.

- TLD nameserver - The top level domain server (TLD) can be thought of as a specific rack of books in a library. This nameserver is the next step in the search for a specific IP address, and it hosts the last portion of a hostname (In example.com, the TLD server is "com").

- Authoritative nameserver - This final nameserver can be thought of as a dictionary on a rack of books, in which a specific name can be translated into its definition. The authoritative nameserver is the last stop in the nameserver query. If the authoritative name server has access to the requested record, it will return the IP address for the requested hostname back to the DNS Recursor (the librarian) that made the initial request.

IP address definition

An IP address is a unique address that identifies a device on the internet or a local network. IP stands for "Internet Protocol," which is the set of rules governing the format of data sent via the internet or local network.

In essence, IP addresses are the identifier that allows information to be sent between devices on a network: they contain location information and make devices accessible for communication. The internet needs a way to differentiate between different computers, routers, and websites. IP addresses provide a way of doing so and form an essential part of how the internet works.

An IP address is a string of numbers separated by periods. IP addresses are expressed as a set of four numbers — an example address might be 192.158.1.38. Each number in the set can range from 0 to 255. So, the full IP addressing range goes from 0.0.0.0 to 255.255.255.255.

IP addresses are not random. They are mathematically produced and allocated by the Internet Assigned Numbers Authority (IANA), a division of the Internet Corporation for Assigned Names and Numbers (ICANN). ICANN is a non-profit organization that was established in the United States in 1998 to help maintain the security of the internet and allow it to be usable by all. Each time anyone registers a domain on the internet, they go through a domain name registrar, who pays a small fee to ICANN to register the domain.

How do IP addresses work

If you want to understand why a particular device is not connecting in the way you would expect or you want to troubleshoot why your network may not be working, it helps understand how IP addresses work.

Internet Protocol works the same way as any other language, by communicating using set guidelines to pass information. All devices find, send, and exchange information with other connected devices using this protocol. By speaking the same language, any computer in any location can talk to one another.

The use of IP addresses typically happens behind the scenes. The process works like this:

1. Your device indirectly connects to the internet by connecting at first to a network connected to the internet, which then grants your device access to the internet.
2. When you are ~~at~~ home, that network will probably be your Internet Service Provider (ISP). At work, it will be your company network.
3. Your IP address is assigned to your device by your ISP.
4. Your internet activity goes through the ISP, and they route it back to you, using your IP address. Since they are giving you access to the internet, it is their role to assign an IP address to your device.
5. However, your IP address can change. For example, turning your modem or router on or off can change it. Or you can contact your ISP, and they can change it for you.
6. When you are out and about – for example, traveling – and you take your device with you, your home IP address does not come with you. This is because you will be using another network (Wi-Fi at a hotel, airport, or coffee shop, etc.) to access

the internet and will be using a different (and temporary) IP address, assigned to you by the ISP of the hotel, airport or coffee shop.

Types of IP addresses

There are different categories of IP addresses, and within each category, different types.

Consumer IP addresses

Every individual or business with an internet service plan ~~will~~ have two types of IP addresses: their private IP addresses and their public IP address. The terms public and private relate to the network location — that is, a private IP address is used inside a network, while a public one is used outside a network.

Private IP addresses

Every device that connects to your internet network has a private IP address. This includes computers, smartphones, and tablets but also any Bluetooth-enabled devices like speakers, printers, or smart TVs. With the growing internet of things, the number of private IP addresses you have at home is probably growing. Your router needs a way to identify these items separately, and many items need a way to recognize each other. Therefore, your router generates private IP addresses that are unique identifiers for each device that differentiate them on the network.

Public IP addresses

A public IP address is the primary address associated with your whole network. While each connected device has its own IP address, they are also included within the main IP address for your network. As described above, your public IP address is provided to your router by your ISP. Typically, ISPs have a large pool of IP addresses that they distribute to their customers. Your public IP address is the address that all the devices outside your internet network will use to recognize your network.

Public IP addresses

Public IP addresses come in two forms – dynamic and static.

Dynamic IP addresses

Dynamic IP addresses change automatically and regularly. ISPs buy a large pool of IP addresses and assign them automatically to their customers. Periodically, they re-assign them and put the older IP addresses back into the pool to be used for other customers. The rationale for this approach is to generate cost savings for the ISP. Automating the regular movement of IP addresses means they don’t have to carry out specific actions to re-establish a customer's IP address if they move home, for example. There are security benefits, too, because a changing IP address makes it harder for criminals to hack into your network interface.

Static IP addresses

In contrast to dynamic IP addresses, static addresses remain consistent. Once the network assigns an IP address, it remains the same. Most individuals and businesses do not need a static IP address, but for businesses that plan to host their own server, it is crucial to have one. This is because a static IP address ensures that websites and email addresses tied to it will have a consistent IP address — vital if you want other devices to be able to find them consistently on the web.

This leads to the next point – which is the two types of website IP addresses.

There are two types of website IP addresses

For website owners who don't host their own server, and instead rely on a web hosting package – which is the case for most websites – there are two types of website IP addresses. These are shared and dedicated.

Shared IP addresses

Websites that rely on shared hosting plans from web hosting providers will typically be one of many websites hosted on the same server. This tends to be the case for individual websites or SME websites, where traffic volumes are manageable, and the sites themselves are limited in terms of the number of pages, etc. Websites hosted in this way will have shared IP addresses.

Dedicated IP addresses

Some web hosting plans have the option to purchase a dedicated IP address (or addresses). This can make obtaining an SSL certificate easier and allows you to run your own File Transfer Protocol (FTP) server. This makes it easier to share and transfer files with multiple people within an organization and allow anonymous FTP sharing options. A dedicated IP address also allows you to access your website using the IP address alone rather than the domain name — useful if you want to build and test it before registering your domain.

Network Classes

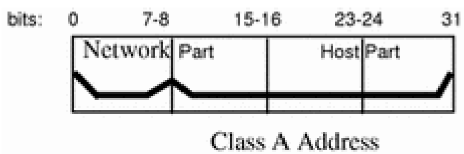
The first step in planning for IP addressing on your network is to determine which network class is appropriate for your network. After you have done this, you can take the crucial second step: obtain the network number from the InterNIC addressing authority.

Currently there are three classes of TCP/IP networks. Each class uses the 32-bit IP address space differently, providing more or fewer bits for the network part of the address. These classes are class A, class B, and class C.

Class A Network Numbers

A class A network number uses the first eight bits of the IP address as its "network part." The remaining 24 bits comprise the host part of the IP address, as illustrated in Figure 3-2 below.

Figure 3-2 Byte Assignment in a Class A Address

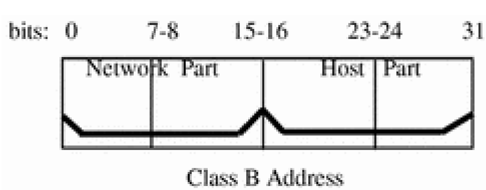


The values assigned to the first byte of class A network numbers fall within the range 0-127. Consider the IP address 75.4.10.4. The value 75 in the first byte indicates that the host is on a class A network. The remaining bytes, 4.10.4, establish the host address. The InterNIC assigns only the first byte of a class A number. Use of the remaining three bytes is left to the discretion of the owner of the network number. Only 127 class A networks can exist. Each one of these numbers can accommodate up to 16,777,214 hosts.

Class B Network Numbers

A class B network number uses 16 bits for the network number and 16 bits for host numbers. The first byte of a class B network number is in the range 128-191. In the number 129.144.50.56, the first two bytes, 129.144, are assigned by the InterNIC, and comprise the network address. The last two bytes, 50.56, make up the host address, and are assigned at the discretion of the owner of the network number. Figure 3-3 graphically illustrates a class B address.

Figure 3-3 Byte Assignment in a Class B Address

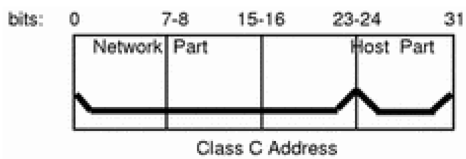


Class B is typically assigned to organizations with many hosts on their networks.

Class C Network Numbers

Class C network numbers use 24 bits for the network number and 8 bits for host numbers. Class C network numbers are appropriate for networks with few hosts--the maximum being 254. A class C network number occupies the first three bytes of an IP address. Only the fourth byte is assigned at the discretion of the network owners. Figure 3-4 graphically represents the bytes in a class C address.

Figure 3-4 Byte Assignment in a Class C Address



The first byte of a class C network number covers the range 192-223. The second and third each cover the range 1- 255. A typical class C address might be 192.5.2.5. The first three bytes, 192.5.2, form the network number. The final byte in this example, 5, is the host number.

Administering Network Numbers

If your organization has been assigned more than one network number, or uses subnets, appoint a centralized authority within your organization to assign network numbers. That authority should maintain control of a pool of assigned network numbers, assigning network, subnet, and host numbers as required. To prevent problems, make sure that duplicate or random network numbers do not exist in your organization.

Designing Your IP Addressing Scheme

After you have received your network number, you can then plan how you will assign the host parts of the IP address.

Table 3-1 shows the division of the IP address space into network and host address spaces. For each class, "range" specifies the range of decimal values for the first byte of the network number. "Network address" indicates the number of bytes of the IP address that are dedicated to the network part of the address, with each byte represented by *xxx*. "Host address" indicates the number of bytes dedicated to the host part of the address. For example, in a class A network address, the first byte is dedicated to the network, and the last three are dedicated to the host. The opposite is true for a class C network.

Table 3-1 Division of IP Address Space

Class	Range	Network Address	Host Address
A	0-127	*xxx*	*xxx.xxx.xxx*
B	128-191	*xxx.xxx*	*xxx.xxx*
C	192-223	*xxx.xxx.xxx*	*xxx*

The numbers in the first byte of the IP address define whether the network is class A, B, or C and are always assigned by the InterNIC. The remaining three bytes have a range from 0-255. The numbers 0 and 255 are reserved; you can assign the numbers 1-254 to each byte **depending on the network number assigned to you**.

Table 3-2 shows which bytes of the IP address are assigned to you and the range of numbers within each byte that are available for you to assign to your hosts.

Table 3-2 Range of Available Numbers

Network Class	Byte 1 Range	Byte 2 Range	Byte 3 Range	Byte 4 Range
A	0-127	1-254	1-254	1-254
B	128-191	Preassigned by Internet	1-254	1-254
C	192-223	Preassigned by Internet	Preassigned by Internet	1-254

Class D networks, which cover the 224.0. 0.0 - 239.255. 255.255 IP address range, are reserved for multicasting,

Class E (240.0. 0.0 - 255.255.255.255) reserved for future experiment and research purpose.

What is IP?

Here, IP stands for **internet protocol**. It is a protocol defined in the TCP/IP model used for sending the packets from source to destination. The main task of IP is to deliver the packets from source to the destination based on the IP addresses available in the packet headers. IP defines the packet structure that hides the data which is to be delivered as well as the addressing method that labels the datagram with a source and destination information.

An IP protocol provides the connectionless service, which is accompanied by two transport protocols, i.e., TCP/IP and UDP/IP, so internet protocol is also known as TCP/IP or UDP/IP.

The first version of IP (Internet Protocol) was IPv4. After IPv4, IPv6 came into the market, which has been increasingly used on the public internet since 2006.

Function

The main function of the internet protocol is to provide addressing to the hosts, encapsulating the data into a packet structure, and routing the data from source to the destination across one or more IP networks. In order to achieve these functionalities, internet protocol provides two major things which are given below.

An internet protocol defines two things:

- Format of IP packet
- IP Addressing system

What is Transmission Control Protocol (TCP)?

Transmission Control Protocol (TCP) is a standard that defines how to establish and maintain a network conversation by which applications can exchange data.

TCP works with the Internet Protocol (IP), which defines how computers send packets of data to each other. Together, TCP and IP are the basic rules that define the internet. The Internet Engineering Task Force (IETF) defines TCP in the Request for Comment (RFC) standards document number 793.

How Transmission Control Protocol works

TCP is a connection-oriented protocol, which means a connection is established and maintained until the applications at each end have finished exchanging messages.

TCP performs the following actions:

- determines how to break application data into packets that networks can deliver;
- sends packets to, and accepts packets from, the network layer;
- manages flow control;
- handles retransmission of dropped or garbled packets, as it's meant to provide error-free data transmission; and
- acknowledges all packets that arrive.

In the Open Systems Interconnection (OSI) communication model, TCP covers parts of Layer 4, the transport layer, and parts of Layer 5, the session layer.

HTTP stands for **H**yper **T**ext **T**ransfer **P**rotocol

WWW is about communication between web **clients** and **servers**

Communication between client computers and web servers is done by sending **HTTP Requests** and receiving **HTTP Responses**

The Hypertext Transfer Protocol (HTTP) is an application-level protocol for distributed, collaborative, hypermedia information systems. This is the foundation for data communication for the World Wide Web (i.e. internet) since 1990. HTTP is a generic and stateless protocol which can be used for other purposes as well using extensions of its request methods, error codes, and headers.

Basically, HTTP is a TCP/IP based communication protocol, that is used to deliver data (HTML files, image files, query results, etc.) on the World Wide Web. The default port is TCP 80, but other ports can be used as well. It provides a standardized way for computers to communicate with each other. HTTP specification specifies how clients' request data will be constructed and sent to the server, and how the servers respond to these requests.

Basic Features

There are three basic features that make HTTP a simple but powerful protocol:

- **HTTP is connectionless:** The HTTP client, i.e., a browser initiates an HTTP request and after a request is made, the client waits for the response. The server processes the request and sends a response back after which client disconnect the connection. So client and server knows about each other during current request and response only. Further requests are made on new connection like client and server are new to each other.
- **HTTP is media independent:** It means, any type of data can be sent by HTTP as long as both the client and the server know how to handle the data content. It is required for the client as well as the server to specify the content type using appropriate MIME-type.
- **HTTP is stateless:** As mentioned above, HTTP is connectionless and it is a direct result of HTTP being a stateless protocol. The server and client are aware of each other only during a current request. Afterwards, both of them forget about each other. Due to this nature of the protocol, neither the client nor the browser can retain information between different requests across the web pages.

FTP

- FTP stands for File transfer protocol.
- FTP is a standard internet protocol provided by TCP/IP used for transmitting the files from one host to another.
- It is mainly used for transferring the web page files from their creator to the computer that acts as a server for other computers on the internet.
- It is also used for downloading the files to computer from other servers.

Objectives of FTP

- It provides the sharing of files.
- It is used to encourage the use of remote computers.
- It transfers the data more reliably and efficiently.

Why FTP?

Although transferring files from one system to another is very simple and straightforward, but sometimes it can cause problems. For example, two systems may have different file conventions. Two systems may have different ways to represent text and data. Two systems may have different directory structures. FTP protocol overcomes these problems by establishing two connections between hosts. One connection is used for data transfer, and another connection is used for the control connection.

Advantages of FTP:

- **Speed:** One of the biggest advantages of FTP is speed. The FTP is one of the fastest way to transfer the files from one computer to another computer.
- **Efficient:** It is more efficient as we do not need to complete all the operations to get the entire file.
- **Security:** To access the FTP server, we need to login with the username and password. Therefore, we can say that FTP is more secure.
- **Back & forth movement:** FTP allows us to transfer the files back and forth. Suppose you are a manager of the company, you send some information to all the employees, and they all send information back on the same server.

Disadvantages of FTP:

- The standard requirement of the industry is that all the FTP transmissions should be encrypted. However, not all the FTP providers are equal and not all the providers offer encryption. So, we will have to look out for the FTP providers that provides encryption.
- FTP serves two operations, i.e., to send and receive large files on a network. However, the size limit of the file is 2GB that can be sent. It also doesn't allow you to run simultaneous transfers to multiple receivers.
- Passwords and file contents are sent in clear text that allows unwanted eavesdropping. So, it is quite possible that attackers can carry out the brute force attack by trying to guess the FTP password.
- It is not compatible with every system.

SMTP

- SMTP stands for Simple Mail Transfer Protocol.

- SMTP is a set of communication guidelines that allow software to transmit an electronic mail over the internet is called **Simple Mail Transfer Protocol**.
- It is a program used for sending messages to other computer users based on e-mail addresses.
- It provides a mail exchange between users on the same or different computers, and it also supports:
    - It can send a single message to one or more recipients.
    - Sending message can include text, voice, video or graphics.
    - It can also send the messages on networks outside the internet.
- The main purpose of SMTP is used to set up communication rules between servers. The servers have a way of identifying themselves and announcing what kind of communication they are trying to perform. They also have a way of handling the errors such as incorrect email address. For example, if the recipient address is wrong, then receiving server reply with an error message of some kind.

Working of SMTP

1. **Composition of Mail:** A user sends an e-mail by composing an electronic mail message using a Mail User Agent (MUA). Mail User Agent is a program which is used to send and receive mail. The message contains two parts: body and header. The body is the main part of the message while the header includes information such as the sender and recipient address. The header also includes descriptive information such as the subject of the message. In this case, the message body is like a letter and header is like an envelope that contains the recipient's address.
2. **Submission of Mail:** After composing an email, the mail client then submits the completed e-mail to the SMTP server by using SMTP on TCP port 25.
3. **Delivery of Mail:** E-mail addresses contain two parts: username of the recipient and domain name. For example, vivek@gmail.com, where "vivek" is the username of the recipient and "gmail.com" is the domain name.
If the domain name of the recipient's email address is different from the sender's domain name, then MSA will send the mail to the Mail Transfer Agent (MTA). To relay the email, the MTA will find the target domain. It checks the MX record from Domain Name System to obtain the target domain. The MX record contains the

domain name and IP address of the recipient's domain. Once the record is located, MTA connects to the exchange server to relay the message.

4. **Receipt and Processing of Mail:** Once the incoming message is received, the exchange server delivers it to the incoming server (Mail Delivery Agent) which stores the e-mail where it waits for the user to retrieve it.
5. **Access and Retrieval of Mail:** The stored email in MDA can be retrieved by using MUA (Mail User Agent). MUA can be accessed by using login and password.

What is POP3?

POP3 is abbreviated as **Post Office Protocol**, and the number three stands for "version 3," which is the latest version and the most widely used email protocol on the internet. Similar to IMAP protocol, it is another protocol for receiving emails from remote servers. It also acts as a message accessing agent to retrieve the message from the mail server to the receiver's system. It helps to protect emails from spam and viruses on the internet.

POP3 works on two modes: **Delete mode** and **Keep mode.**

It works on **Delete mode** when a user is accessing an email service from a permanent device. In this, once the mails are downloaded or retrieved from the mailbox, mails are deleted from the mailbox permanently.

It operates on **Keep mode** when the user is not accessing mails from the primary device. In this, it keeps the mails after retrieval also for later retrieval.

When to Use POP3?

Use POP3 for the following cases:

- We should use POP3 if we want to access mails using a single device
- If the number of emails is high.
- If we want to access the mails offline.

Features of POP3

- In POP3, all the emails are downloaded to the local computer, and once all the emails are downloaded, they are deleted from the server.

- Downloaded emails can be accessed offline also.
- Emails are not synchronized between different devices, which means if we set up our email using our mobile phone with POP3, those emails will be downloaded completely on your mobile phone, and cant be accessed from other devices.

The **Telnet Protocol** (**TELNET**) provides a standard method for terminal devices and terminal-oriented processes to interface.

TELNET is commonly used by terminal emulation programs that allow you to log into a remote host. However, **TELNET** can also be used for terminal-to-terminal communication and interprocess communication. **TELNET** is also used by other protocols (for example, **FTP**) for establishing a protocol control channel.

The name "Telnet" is short for "Teletype Network Protocol".
In a nutshell, Telnet is a computer protocol that was built for interacting with remote computers. It enables terminal-to-terminal communication and can be used for a variety of purposes.
The word "Telnet" also refers to the command-line utility "telnet", available under Windows OS and Unix-like systems, including Mac, Linux, and others. We will use the term "Telnet" mainly in the context of the telnet client software.
Telnet utility allows remote user access to test connectivity to remote machines and issue commands through a keyboard. Though most users opt to work with graphical interfaces, Telnet is one of the simplest ways to check connectivity on certain ports.

What is Telnet used for?

The Telnet protocol is as old-school as it gets in the tech world. It was developed in 1969, so it doesn't pack the robust security features of today's network protocols.
As we've mentioned, via Telnet, users can connect to software that utilizes text-based unencrypted protocols from web servers to ports. You can open the command line interface on a remote computer, type "telnet", the remote machine's name or IP address, and wait for the Telnet connection to ping the port to check if it's open.
However, you can use Telnet to execute various other tasks. When properly connected, you can edit files, run programs, or check your email with it.

Gopher is an application-layer protocol that provides the ability to extract and view Web documents stored on remote Web servers. Gopher was conceived in 1991 as one of the Internet’s first data/file access protocols to run on top of a TCP/IP network. It was developed at University of Minnesota and is named after the school's mascot.

Gopher was designed to access a Web server or database via the Internet. It requires that files be stored in a menu-style hierarchy on a Gopher server that is accessible through a Gopher-enabled client browser and/or directly. It initially supported only text-based file/document access but later came to support some image formats such as GIF and JPEG.

Gopher was succeeded by the HTTP protocol and now has very few implementations. Gopher-based databases, servers or websites can be accessed through two search engines: Veronica and Jughead.

WAIS (pronounced "ways") was a client-server database search system introduced in 1990 before the widespread adoption of the World Wide Web. It indexed the contents of databases located on multiple servers and made them searchable over networks, including the Internet. Text searches of the databases were ranked by relevance, where files with more keyword hits are listed first.

As the web grew in popularity, WAIS usage declined and eventually faded into obsolescence. The service has now been completely replaced by modern search engines.

How WAIS Functioned

A central server called the Directory of Servers stored and indexed a database of other publicly-available databases. Since data storage was limited by the technology of the day, databases were distributed amongst several servers. To conduct a WAIS search, one would connect via telnet to a server running a WAIS client or install and run their own client. Once connected to the Directory of Servers, a WAIS user could browse a large number of distributed databases, each focused on a specific topic (usually an academic discipline, such as history, biology, or social sciences).

WAIS was also popular as a way to search for text within individual Gopher servers. Modern search engines as a way to quickly find relevant information were inspired by WAIS. As the World Wide Web caught on and supplanted Gopher servers, WAIS declined in popularity

1. Internet :

The network formed by the co-operative interconnection of millions of computers, linked together is called Internet. Internet comprises of :

- **People :** People use and develop the network.
- **Resources :** A collection of resources that can be reached from those networks.
- **A setup for collaboration :** It includes the member of the research and educational committees worldwide.

2. Intranet :

It is an internal private network built within an organization using Internet and World Wide Web standards and products that allows employees of an organization to gain access to corporate information.

3. Extranet :

It is the type of network that allows users from outside to access the Intranet of an organization.

Difference between Internet, Intranet and Extranet :

Point of difference	**Internet**	**Intranet**	**Extranet**
Accessibility of network	Public	Private	Private
Availability	Global system.	Specific to an organization.	To share information with suppliers and vendors it makes the use of public network.
Coverage	All over the world.	Restricted area upto an organization.	Restricted area upto an organization and some of its stakeholders or so.
Accessibility of content	It is accessible to everyone connected.	It is accessible only to the members of organization.	Accessible only to the members of organization and

Point of difference	**Internet**	**Intranet**	**Extranet**
			external members with logins.
No. of computers connected	It is largest in number of connected devices.	The minimal number of devices are connected.	The connected devices are more comparable with Intranet.
Owner	No one.	Single organization.	Single/ Multiple organization.
Purpose of the network	It’s purpose is to share information throughout the world.	It’s purpose is to share information throughout the organization.	It’s purpose is to share information between members and external, members.
Security	It is dependent on the user of the device connected to network.	It is enforced via firewall.	It is enforced via firewall that separates internet and extranet.
Users	General public.	Employees of the organization.	Employees of the organization which are connected.
Policies behind setup	There is no hard and fast rule for policies.	Policies of the organization are imposed.	Policies of the organization are imposed.
Maintenance	It is maintained by ISP.	It is maintained by CIO. HR or communication department of an organization.	It is maintained by CIO. HR or communication department of an organization.

Point of difference	**Internet**	**Intranet**	**Extranet**
Economical	It is more economical to use.	It is less economical.	It is also less economical.
Relation	It is the network of networks.	It is derived from Internet.	It is derived from Intranet.
Example	What we are normally using is internet.	WIPRO using internal network for its business operations.	DELL and Intel using network for its business operations.

Intranet Advantages & Disadvantages

Advantages

1. They reduce the number of systems your workforce needs.

Matthew Fox of Valiant Technology says, “The workday tends to move quickly, especially with so much technology at our disposal. It seems like we have an infinite number of systems available where we can store information and communicate with our team. Having one system to replace 10 is much easier to deal with.”

He notes that storing commonly accessed and important documents in a single solution makes them easier to find, and much more likely to be referenced. This also reduces delays and interruptions from team members hunting to find the information they need, improves productivity, and can even enable employees to provide a higher level of customer service.

2. They act as a one-stop shop for all things collaboration.

Robust intranets facilitate multiple facets of collaboration:

- Sharing important documents for internal and external projects.
- Creating customised spaces for specific purposes, such as departmental happenings or client engagements.
- Enabling task management so projects can stay on track.

Fox shares a story about how a former manager used to call their intranet a "one-stop shop" because it had everything employees needed to perform their day-to-day duties. Staff had easy access to the things they needed. When a new need was identified, they would collaborate on a way to evolve the intranet to address it.

"Our intranet helped us easily work together to make things happen instead of running around in disorganised fashion. It made everyone a stakeholder, ultimately changing the way we organised projects and solved problems, and freed up time for activities that helped the business grow," explains Fox.

3. They support your company culture.

Company culture permeates every aspect of the workplace (even in hybrid or remote arrangements), from employee interactions to management styles to the tools you use. **Using an intranet to, say, onboard new employees gives you the opportunity to shape how they see the organisation.** *(Tweet this!)* Another example: You can share stories with the whole company about how employees are embracing company values on customer projects.

Sergey Yudovskiy of electroNeek says an intranet is a place for your corporate culture to shine: "You can share ideas, congratulate others on their successes, and connect better by sharing personal news. A well-organised intranet can become a hive of activity that furthers the company culture with every employee."

Disadvantages

1. They can take a lot of time and effort to implement.

Most modern intranet solutions are easy to implement and use; however, Sergey Korolev of Itransition says that depending on the solution you choose, implementing an intranet can take lots of organisational effort and investment, especially if a company chooses custom development. "Customisations can take a year or more, and there still might be no explicit benefits from the implemented solution if not planned and used correctly."

2. They can require significant oversight.

Fox says that without the proper administration, an intranet can tend to become disorganised. "Organisations typically choose intranets to help keep things organised and ensure employees have a solid tool for collaboration. The ease of use of today's intranets makes *choosing* an intranet easy; however, some users can go overboard."

Fox explains that a key feature of modern intranets is the ability to build mini-intranets, or individualised spaces within the overall intranet structure. "Picture a fenced-in garden as your intranet. You may have a section for roses that's separated by rocks and a number of other walled-off sections for other plants—these are the mini-intranets that represent different departments or smaller teams."

The problem, he notes, is that some organisations that employ these mini-intranets wind up suffering from disorganisation and siloed information—the very things they wanted to avoid by using an intranet. "When this happens, business operations suffer, not to mention the unnecessary investment."

To avoid this, Korolev recommends taking a systematic approach to building each space, and employing clear content management guidelines for the overall intranet and the spaces within. “Otherwise, intranets can quickly become messy, with users having to spend a lot of unnecessary time finding the information they need or accomplishing what should be simple tasks.”

3. They can have low user adoption.

Sergey Golubenko of ScienceSoft says many intranets unfortunately go unused by the very employees they’re meant to help. The main reasons are because intranets often have too many confusing features, customizations, and integrations. In addition, he notes some intranets are plagued with complicated navigation and ineffective search—there’s nothing worse than not being able to find what you need when you’re strapped for time. These challenges make intranets less appealing to employees who want the easiest route to handle their job duties.

“Overall, intranets on the market tend to lack a user-friendly design,” Golubenko explains. “Along with the lack of proper training, this leads to a situation where the majority of employees rarely visit their workplace intranet, if they use it at all.”

When you’re searching the market for an intranet, be sure to consider its design. How easy is it to navigate? How quickly can you figure out its features? You can even try a few test runs with a small team—they can use each intranet as part of their daily routine and report back on their experiences. This way, you’ll have a better idea of how your organisation will receive the new solution.

Extranet Advantages & Disadvantages

Advantages

1. They improve communication with external parties.

Today's business operations often require just as much collaboration with clients and vendors as internal teams. Extranets help you keep your work with these outside parties organised, so nothing gets lost.

For example, you may have a large undertaking for a client project that requires input from a vendor in terms of supplies. Instead of communicating and sharing files with each entity separately, you can bring them together in an extranet to keep all related work in one place, with the ability to restrict access to files to specific groups or individuals.

2. They help you in the same ways an intranet does.

Intranets and extranets are similar, but Fox explains that today's cloud-based intranet solutions have replaced the need for pure extranets in many scenarios. "You can typically restrict the access of third parties to specific spaces or items, which makes it easy to share information across internal systems without giving clients or vendors free rein. You can ensure both internal and external parties can find the information they need in the same environment."

With this integration, an extranet takes on the mantle of intranet, providing the same advantages of an intranet that were covered in the list above.

Find more reasons to get started using online extranet today in this article.

Disadvantages

1. They can require ongoing maintenance and attention.

Golubenko says that traditional extranets often require regular maintenance to remain effective, including functional and security updates. “Such maintenance demands an in-house IT team or outsourced IT services, which also incurs significant expenses.”

He notes that an extranet also requires constant content governance to avoid becoming overloaded with irrelevant and unorganised content, and presenting information that is either erroneous or intended for another client or vendor.

2. They aren’t always appealing to users.

Golubenko says most extranets have too many features, customisations, and integrations, making them not very user-friendly. “The result is overly complicated navigation, ineffective search, and an overwhelming selection of widgets and apps. Along with the lack of proper training, this leads to a situation where the majority of users rarely visit an extranet, or don’t use it at all.”

Korolev adds another perspective, explaining that it’s difficult to get all external parties to use an extranet, especially if they are accustomed to other solutions. “Clients especially may be already using their own internal systems, and they won’t have the same pressure as employees for using the solution.”

3. They can pose a higher security risk.

“With extranets, you’re opening up the digital walls of your company to people outside your organisation,” says Golubenko. Whether it’s clients or suppliers, these individuals are not motivated or bound in the same way as your employees. Granting them access to your system raises additional security concerns for you to consider.

To control this risk, it’s important for your extranet solution to have in place advanced security measures, such as SSL encryption. More importantly, it should have clear user access measures so you can control the information external parties are able to view and change. What’s even more helpful is the capability to build dedicated workspaces that can compartmentalise internal operations and projects that require external collaboration.

Advantages of Internet

Internet technology has numerous advantages. You can connect with anyone in this world without facing any errors. The internet has brought the world close to each other. Have a look at some essential **benefits of internet**.

1. Source of Information
2. Source of Entertainment
3. Keep Informed
4. Online Shopping
5. Communication
6. Social Networking
7. Learning
8. Earn Money from the Internet
9. Online Banking
10. Accessibility

Disadvantages of Internet

The internet has many conveniences, but excessive usage of the internet can affect you in a variety of ways. There are a lot of uses and limitations of the internet. So, let's explore the **drawbacks of the internet.**

1. A Waste of time
2. Not Safe Place for Children
3. Privacy Exposure
4. Money Frauds
5. Virus/Malware
6. Online Threatening or Harassment
7. Cyberbullying
8. Wrong Information
9. Health Issues
10. Stalking

World Wide Web

World Wide Web (Web) or WWW is an area of information used by global identifiers (URLs / Uniform Resource Locators) to identify useful resources.

WWW is a collection of server sites from around the world with the aim of providing data and information for the exchange.

Different between Internet and World Wide Web.

Some people usually use the words “Internet” and “World Wide Web” interchangeably. They think they’re the same thing, but in fact it’s completely different. The Internet is completely different from the WWW. The Internet is a global network of devices such as computers, laptops, etc. The Internet enables people to send emails to other people and chat with them online. For example, when you send an email or chat with someone online, you’re using the Internet.

However, if you want to open a website like google.com for search, use the World Wide Web. a network of servers over the Internet. You request a web page from your computer using a browser, and the server renders that page in your browser.

The computer is called a client, which runs a program (web browser) and asks the other server for the required information.

The Evolution of the World Wide Web

Web 1.0 – Read Only Web

Web 1.0 was the first web generation that is a new revolution in the world of the Internet because it has changed the way the industry and the media work. Basically, the website that was built in the first generation is generally developed to someone who want to accessing information and it has a slightly interactive nature where visitors can only read because there is no process to input data.

Web 2.0 – Read and Write Web

Web 2.0 was the second stage of the World Wide Web. This generation was also known as the "The Social web" because in this generation the users not only able to read the websites but they also can connect with other users. Now, most of the people started to collaborated on ideas, they could share information and create information that was available to the whole world.

Growth of IoT or Internet of Things

In web 2.0, the focus was on connection and interaction between people but Web 3.0 focuses is mainly on connecting people with devices using the internet, that's why we call it "Internet of Things".

IoT means the interconnection between all of the devices and the internet so that they can send and receive data. It is most important developments in the field of internet and will finally lead from Web 3.0 to Web 4.0. Now we can see several mobile apps based on IoT which connect many homes or office devices that interact with each other through the internet.

Web 3.0 – The Semantic Web

Web 3.0 is a technologies that give a new way that for helping the computer to organize and draw conclusions from the data online. By definition, basically the Semantic Web has the same goal because the Semantic Web has Web content that can not only be expressed in natural language understood by humans, but also in forms that can be understood, interpreted and used by software agents. Through this Semantic Web, various software will be able to search, share and integrate information in an easier or more precise way Web 3.0 is a realization of the development of artificial intelligence systems (artificial intelligence) to create global data that can be understood by the system, so the system can reinterpret these data to visitors properly.

Web 4.0 – The Symbiotic Web

Web 4.0 is also known as symbiotic web. The dream behind of the symbiotic web is interaction between humans and machines in symbiosis. It will be possible to build more powerful interfaces such as mind controlled interfaces using web 4.0. In simple words, machines would be clever on reading the contents of the web.

Web 4.0 will be the read-write-execution-concurrency web. Web 4.0 (The webOS) will be like a middleware in which will functioning like an operating system. The webOS will be parallel to the human brain and implies a massive web of highly intelligent interactions. Although there is no exact idea about web 4.0 and its technologies, but it is obvious that the web is moving toward using artificial intelligence to become as an intelligent web.

What is web architecture?

Web architecture is the process of designing, creating and implementing an internet-based computer program. Often, these programs are websites that contain useful information for a user, and web developers may design these programs for a particular purpose, company or brand. Web architecture entails every component of an application and helps web developers create designs that enhance a user's experience. Ideally, a completed project allows users to access information easily and understand how to navigate its content. While there are many different approaches, strategies and patterns of web application architecture, the process commonly influences these components:

- **Design:** This refers to how the program appears aesthetically to a user.
- **IT infrastructure:** This element of a web architecture program includes coding and software development involved in designing a program.
- **User experience:** Besides design aspects, web architecture concepts allow developers to implement other program functions that appeal to a user, such as simple language or easy navigation.
- **Software:** Effective web architecture often includes the development of robust software to support the functions of a program or application.
- **Monetization:** This refers to the ability to monetize a web program, which developers incorporate into the foundational architecture of a program.
- **Efficiency:** The methods used to create a program directly relate to its efficiency—both in its design and at a software level.
- **Reliability:** This refers to the web application's trustworthiness and consistency when a user attempts to access it. Emphasizing this component can help developers avoid technical glitches and other issues that may arise on a user's end.
- **Scalability:** While designing, it's important for web developers to consider the potential need to eventually increase the size and reach of the program.

- **Security:** This refers to the level of network security a web program possesses to ensure the protection of user data.

Why is web architecture important?

Web architecture is important because it functions as the foundation for each element of a digital application or program. Since it's foundational, the quality of the architecture often directly impacts the quality of the finished product. Additionally, effective web architecture can ensure adaptability if an employer requests major changes to a project that's already underway. It's important to develop a secure foundation of web architecture to ensure that a project can progress efficiently and within an established budget.

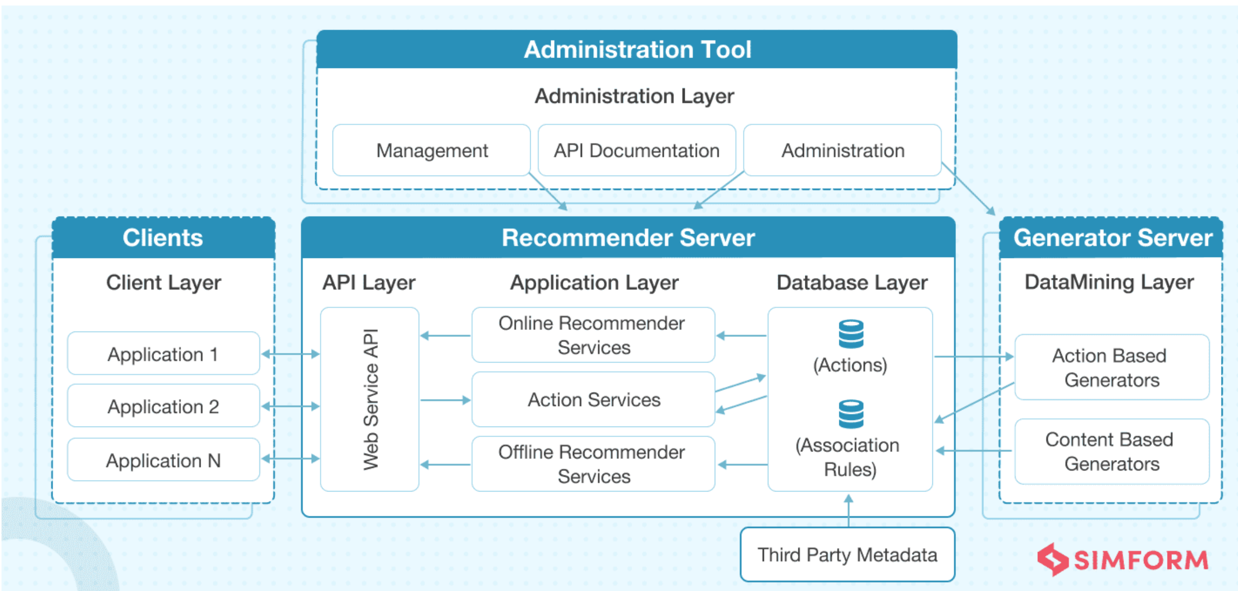
Web architecture is becoming more relevant as the use of technology- and internet-based programs grows. The use of the internet to conduct business—both locally and globally—has increased the need for high-functioning web programs and websites. Other benefits of web architecture include:

- **Increased network security:** The overall security of a web program depends on its infrastructure. Effective web architecture can secure users' data and ensure the reliability of the program itself.
- **Program stability:** A solid infrastructure also allows multiple users to access a program while remaining stable and functional for each user.
- **Fast processing of user requests:** Clean code and well-developed programs are likely to process requests from a user with more speed and efficiency. This can enhance the user's experience and benefit the usability of the program.
- **Ability to collect data:** Web and software developers often collect data, such as how many users visited a particular page or users' locations, from the site or program they develop. They implement this ability into the architecture so they can use this data to further develop and improve the program for their target audience.
- **Scaling the product:** Another element that web architecture impacts is the ability to scale the program or website. For example, if developers build a program that can originally host 10,000 visitors, web architecture can ensure operability as more users continue to join past that threshold.

What is Web Application Architecture?

Web application architecture is a blueprint of simultaneous interactions between components, databases, middleware systems, user interfaces, and servers in an application. It can also be described as the layout that logically defines the connection between the server and client-side for a better web experience.

Web Application Architecture Diagram



Why is Web App Architecture Important?

Market trends keep changing, user expectations keep evolving, and the growth of a business is never-ending. A web app needs an architecture to lay a strong foundation, and without it, your business app will be diving in the big ball of mud architecture anti-pattern.

A well-thought web app architecture can handle the various loads and adapt to the changing business requirement skillfully to deliver a fast user experience that further improves the app performance. You can also take on several development tasks at the same time by dividing the structure into several small modules, eventually reducing the development time as well. Furthermore, it becomes easier to integrate new functionalities without affecting the overall structure.

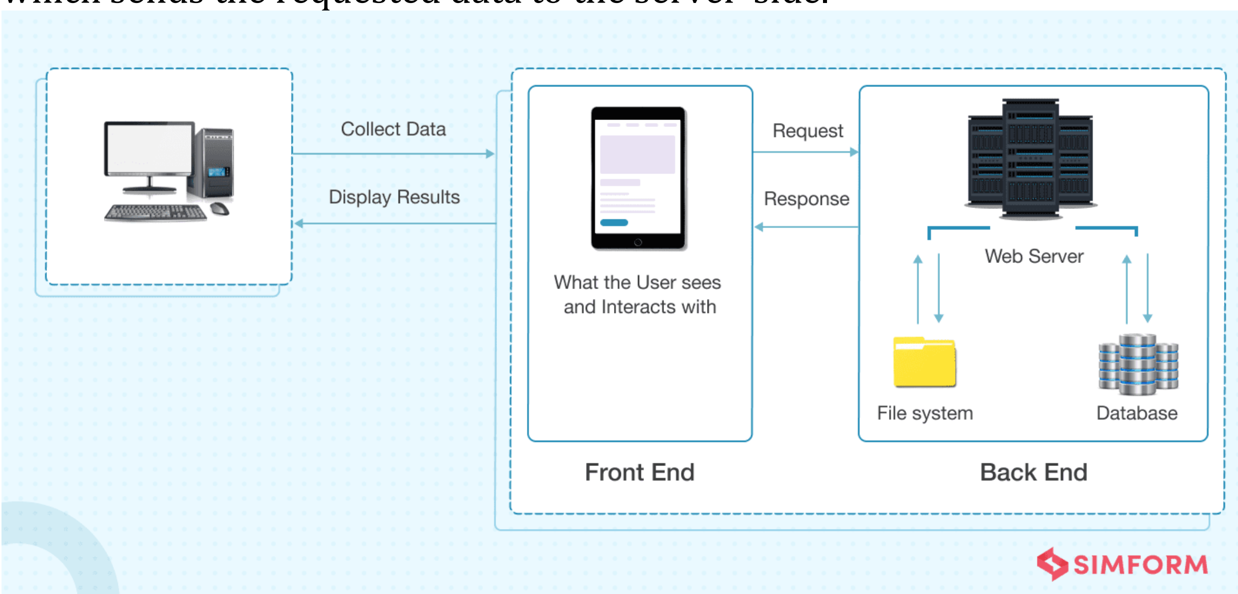
And how about security you may ask. Well, web architecture divides the application into several blocks that are secured separately to minimize security threats, including the risk posed by malicious codes. Apart from that, applications with a future-proof architecture provide the opportunity to add new features and maintain low latency, even when user count increases.

How does the Web App Architecture Work?

All applications are made up of two primary components:

- **Client-side**, popularly called**:** the frontend, where the code is written in HTML, CSS, JavaScript and stored within the browser. It's where user interaction takes place.
- **Server-side**, also known as the backend, controls the business logic and responds to HTTP requests. The server-side code is written in Java, PHP, Ruby, Python, etc.

Apart from this, there is an additional component i.e. database server, which sends the requested data to the server-side.



Let's understand how an architecture functions:

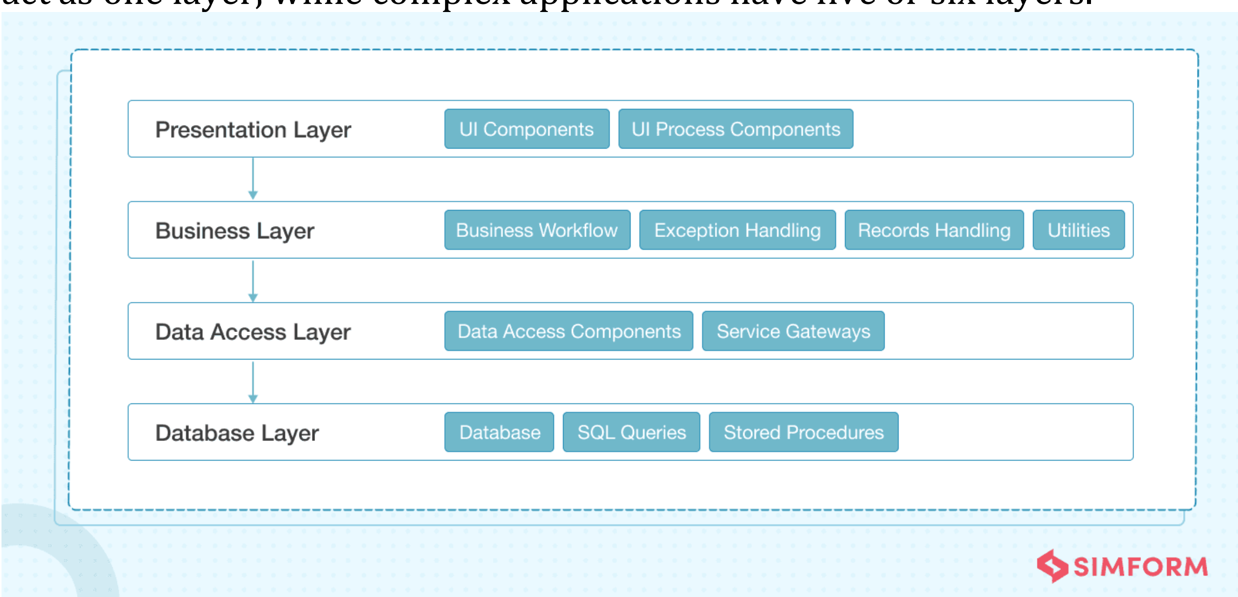
You type a URL, for example, 'walmart.com' in the browser and hit enter. The browser will send a request to the Domain Name Server that will recognize the IP address and further send your request to the server where Walmart is located. The server then catches the request and sends it to the data storage to locate the page and requests for data to be displayed on the browser. The page is then displayed on your screen with the requested information.

Find out how you can build scalable web applications?

TIPS FROM THE EXPERTS

Layers of Web App Architecture

Modern web applications follow a layered architecture that consists of presentation, business, persistence, and database layers. Small applications have three layers, where, in some cases, the business and persistence layers act as one layer, while complex applications have five or six layers.



1. **Presentation layer**, built with HTML, CSS, JavaScript, and its frameworks, enables communication between the interface and browser to facilitate user interaction.
2. **Business layer** defines the business logic and rules. It processes browser requests, executes the business logic

associated with the requests, and then sends it to the presentation layer.

3. **Persistence layer** is responsible for data persistence and is also known as Data Access Layer. It's closely connected to the business layer and has a database server that retrieves data from corresponding servers.
4. **Database layer**, also known as Data Service Layer, holds all the data and ensures data security by separating the business logic from the client-side.

Each of these layers works in isolation. That means each layer works independently. Components of one layer are closed and deal with the associated layer's logics. For example, components residing in the presentation layer deal with presentation logic, while the components in the business layer deal with business logic.

It also reduces the future burden while changes are required in the web application. Hence, changes can be made in one layer without affecting the components of other layers.

Web Application Components

Web app components are broadly divided into two parts–

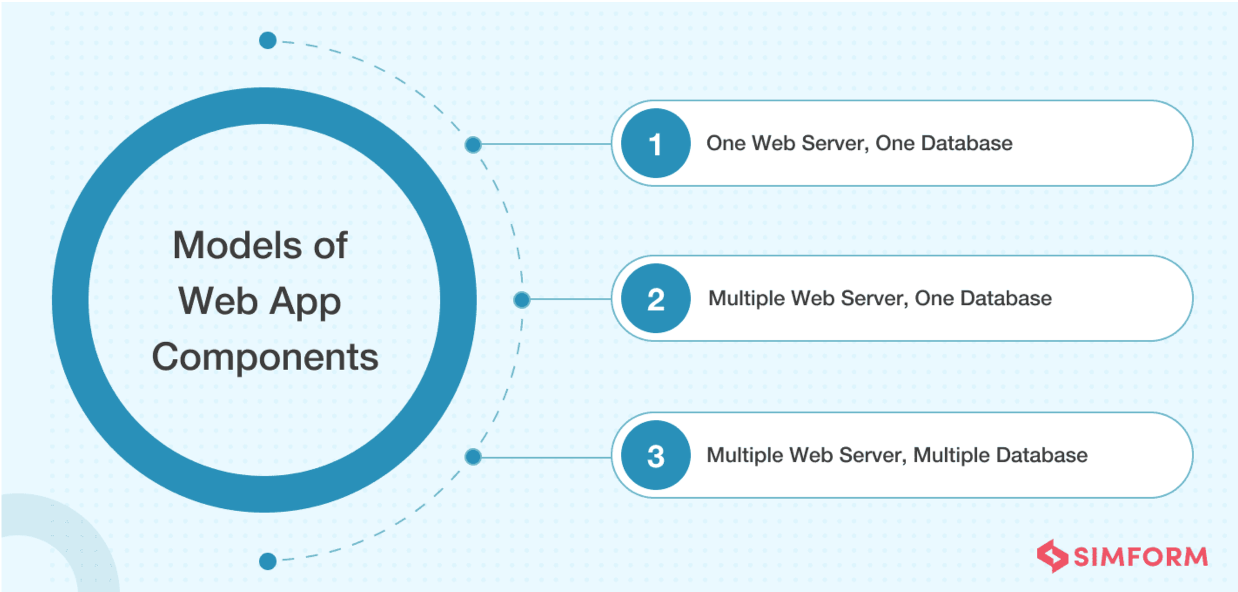
User interface components are a part of the visual interface of a web application and have no interaction with the architecture. They're restricted to a web page's display and consist of activity logs, configuration settings, dashboards, statistics, widgets, notifications, etc., to enhance the user experience.

Structural web components consist of client and server components. Client components exist in the user's browser and interact with the functionality of web applications. HTML, CSS, and JavaScript are commonly used for building these components.

On the other side, server components are bifurcated into a web app server that handles business logic and a database server that stores data. PHP, JAVA, Python, Node.js, .NET, and Ruby on Rails are some frameworks used for creating server components.

Models of Web Applications

There are several models used to build these aforementioned components. Simform always believes in opting for the best model to serve your business purpose with excellent app performance. Our team of engineers is skilled in structuring the best architecture model according to the **type of web application** you want to build. Let's take a look at all the available options:



One web server, one database model might be a little outdated as it has only one server and database to handle all requests. This means if the server goes down so will your app. However, it is generally used for test practices and is a good option if you are a startup having budgetary constraints.

Multiple web servers, one database model reduces the risk of data failure as a backup server is always available if one server goes down. However, the chances of the website crash may still exist due to the availability of only one database.

Multiple web servers, multiple databases model reduces the app's performance risk since there are two options for database storage. You can either store identical data on all the databases or distribute it evenly among all servers.

Types of Web Application Architecture

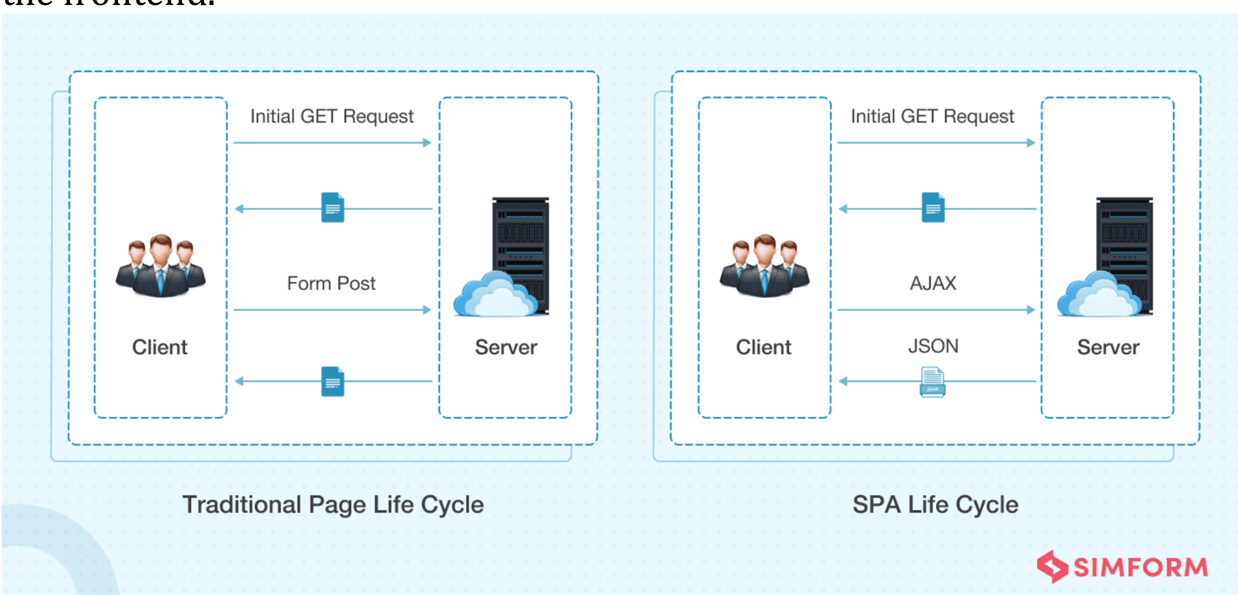
It is always a good practice to select the most appropriate architecture considering various factors in mind, such as app logic, features, functionalities, business requirements, etc. The right architecture defines the purpose of your product as a whole.

Web applications are broadly divided into four types, as listed below:

Single Page Application Architecture

SPA (Single Page Applications) has been introduced to overcome the traditional limitations to achieve smooth app performance, intuitive, and interactive user experience.

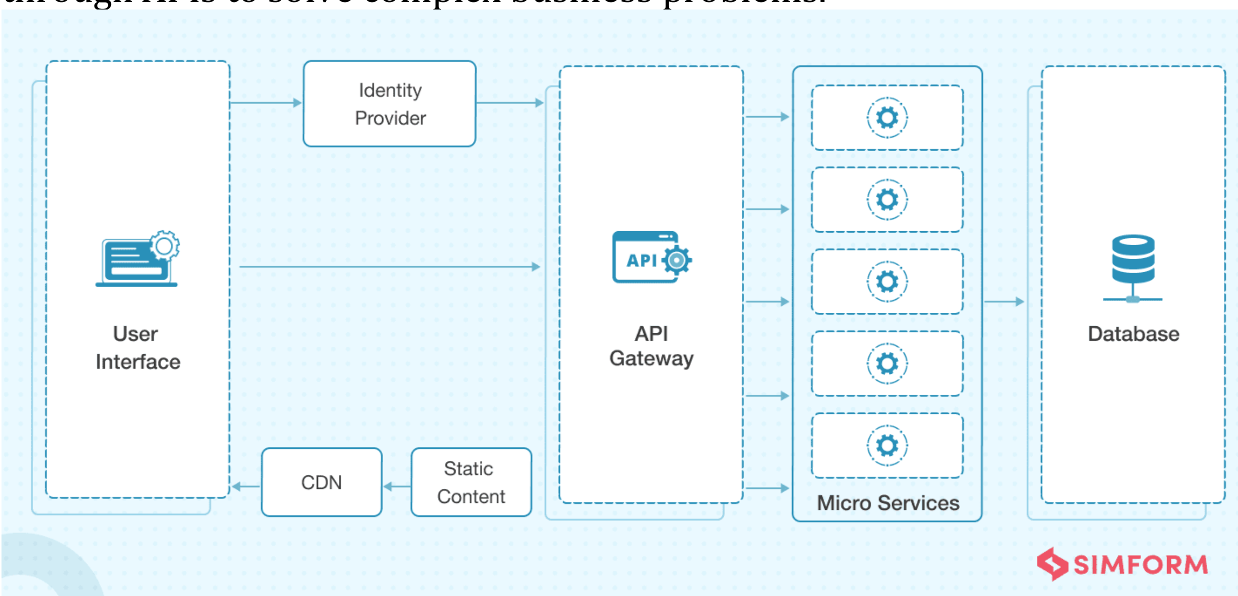
Instead of loading a new page, SPAs load a single web page and reload the requested data on the same page with dynamically updated content. The rest of the web page remains untouched. They are developed on the client-side **using JavaScript frameworks** as the entire logic is always shifted to the frontend.



We worked with **Food Truck Spaces** to build a tech solution that could act as an automated directory between event organizers, property owners, and food entrepreneurs. We developed a single-page web application using AngularJS and ASP.Net API to automate the process of booking, advertising, and online transactions.

Microservice Architecture

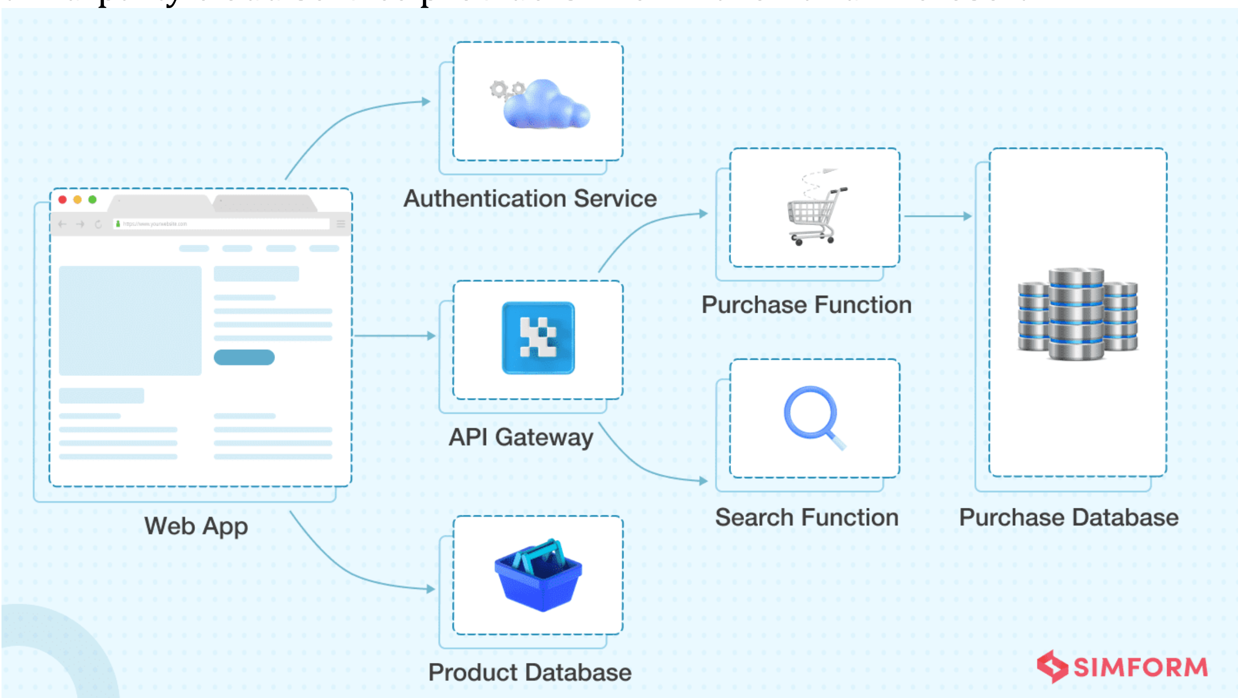
Microservice architecture has become the best alternative to Service-Oriented Architecture (SOA) and monolithic architecture. The services are loosely coupled to be developed, tested, maintained, and deployed independently. Such services can also communicate with other services through APIs to solve complex business problems.



Deployment of web apps using monolithic architecture is a cumbersome task because of its tightly coupled components. Microservices has resolved this issue by separating the application into multiple individual service components. It further simplifies the connectivity between service components and eliminates the need for service orchestration. Some tech giants who are popular for using microservices are Amazon, Netflix, SoundCloud, Comcast, and eBay.

Serverless Architecture

It’s an architecture where the whole execution of code is taken care of by cloud service providers– no need to deploy them manually on your server. Serverless architecture is a design pattern where applications are built and run without any manual intervention on the servers that are managed by third-party cloud service providers like Amazon and Microsoft.

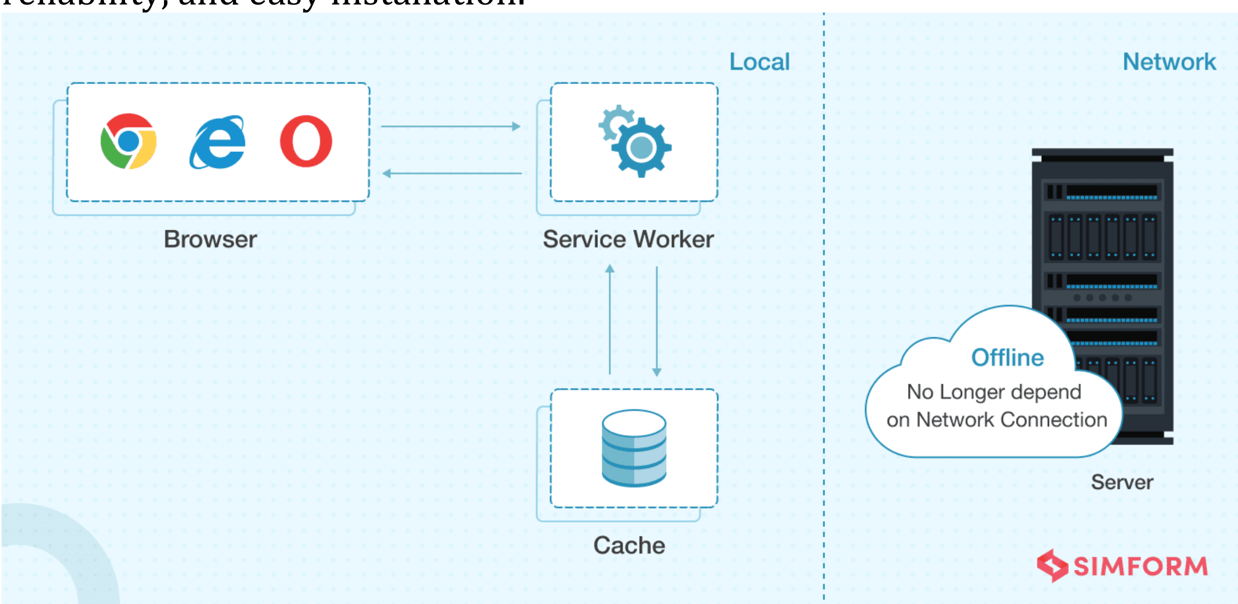


It lets you focus more on the quality of the product and complexity to make them highly scalable and reliable. Broadly, it is categorized into two types – Backend-as-a-Service (BaaS) and Function-as-a-Service (FaaS).
BaaS lets developers focus on the frontend development tasks, eliminating the operations performed on the backend. AWS Amplify, for one, is a popular BaaS product. FaaS, on the other hand, is an event-driven model that allows developers to break the applications into small functions to

focus on the code and event triggers. The rest will be handled by the FaaS service providers such as AWS Lambda and Microsoft Azure.
We partnered with **Fédération Internationale de Hockey (FIH)** to develop a highly scalable and interactive web app. Tech stacks like **headless CMS**, React, and integrated APIs were used to support the serverless features. Our experts also leveraged Amazon EC2 to handle the colossal traffic, Amazon S3 to store videos, and Amazon CloudFront as a Content Delivery Network (CDN).

Progressive Web Applications

Google introduced Progressive Web Apps (PWAs) in 2015 to develop apps that offer rich and native functionality with enhanced capabilities, reliability, and easy installation.

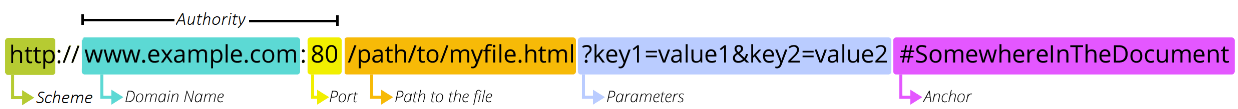


PWAs are compatible apps with any browser and can run on any device. You can easily adjust an app's function to a tablet and a desktop as well. These apps can easily be discovered and shared through URL instead of the app store. Installation of these apps is also effortless and can be quickly added to a device’s home screen. These apps work efficiently on poor internet connectivity and in offline mode as well.

Uber, Aliexpress, Alibaba, Pinterest, and Starbucks are few famous companies who are known for developing their products in the form of PWAs.

URL stands for *Uniform Resource Locator*. A URL is nothing more than the address of a given unique resource on the Web. In theory, each valid URL points to a unique resource. Such resources can be an HTML page, a CSS document, an image, etc. In practice, there are some exceptions, the most common being a URL pointing to a resource that no longer exists or that has moved. As the resource represented by the URL and the URL itself are handled by the Web server, it is up to the owner of the web server to carefully manage that resource and its associated URL.

A URL is composed of different parts, some mandatory and others optional. The most important parts are highlighted on the URL below (details are provided in the following sections):



What is a Browser?

A browser is a software program that is used to explore, retrieve, and display the information available on the World Wide Web. This information may be in the form of pictures, web pages, videos, and other files that all are connected via hyperlinks and categorized with the help of URLs (Uniform Resource Identifiers). For example, you are viewing this page by using a browser.

A browser is a client program as it runs on a user computer or mobile device and contacts the webserver for the information requested by the user. The web server sends the data back to the browser that displays the results on internet supported devices. On behalf of the users, the browser sends requests to web servers all over the internet by using HTTP (Hypertext Transfer Protocol). A browser requires a smartphone, computer, or tablet and internet to work.

A browser is **an application program that provides a way to look at and interact with all the information on the World Wide Web**. This includes Web pages, videos and images.

Features of Web Browser

Most Web browsers offer common features such as:

1. **Refresh button:** Refresh button allows the website to reload the contents of the web pages. Most of the web browsers store local copies of visited pages to enhance the performance by using a caching mechanism. Sometimes, it stops you from seeing the updated information; in this case, by clicking on the refresh button, you can see the updated information.
2. **Stop button:** It is used to cancel the communication of the web browser with the server and stops loading the page content. For example, if any malicious site enters the browser accidentally, it helps to save from it by clicking on the stop button.
3. **Home button:** It provides users the option to bring up the predefined home page of the website.
4. **Web address bar:** It allows the users to enter a web address in the address bar and visit the website.
5. **Tabbed browsing:** It provides users the option to open multiple websites on a single window. It helps users to read different websites at the same time. For example, when you search for anything on the browser, it provides you a list of search results for your query. You can open all the results by right-clicking on each link, staying on the same page.
6. **Bookmarks:** It allows the users to select particular website to save it for the later retrieval of information, which is predefined by the users.

Netscape Navigator was a proprietary web browser, and the original browser of the Netscape line, from versions 1 to 4.08, and 9.x. It was the flagship product of the Netscape Communications Corp and was the dominant web browser in terms of usage share in the 1990s, but by around 2003 its user base had all but disappeared.[2] This was partly because the Netscape Corporation (later purchased by AOL) did not sustain Netscape Navigator's technical innovation in the late 1990s.

Internet Explorer

Internet Explorer is a free web browser, commonly called IE or MSIE, that allows users to view web pages on the internet. It is also used to access online banking, online marketing over the internet, listen to and watch streaming videos, and many more. It was introduced by **Microsoft** in **1995**. It was produced in response to the first geographical browser, **Netscape Navigator**.

Microsoft Internet Explorer was a more popular web browser for many years from 1999 to 2012 as it surpassed the Netscape Navigator during this time. It includes network file sharing, several internet connections, active Scripting, and security settings. It also provides other features such as:

- **Remote administration**
- **Proxy server configuration**
- **VPN and FTP client capabilities**

Opera: An Opera web browser was first conceived at Telenor company in 1994, later bought by the Opera Software on 1 April 1995. It was designed for desktop and mobile interfaces, but it is more popular now for mobile phones. It is based on Chromium, and it uses the blink layout engine. An opera mini was released for smartphones on 10 August 2005 that could run standard web browsers. It can be downloaded from the google play store or Apple play store.

Mozilla Firefox

Mozilla Firefox is an open-source web browser that is used to access the data available on the World Wide Web. As compared to Internet Explorer, the popular Web browser Firefox provides users a simple user interface and faster download speeds. It uses the **Gecko layout engine** to translate web pages, which executes current and predicted web standards.

Firefox was widely used as an alternative to Internet Explorer 6.0 as it provided user protection against spyware and malicious websites. In the year of 2017, it was the fourth-most widely used web browser after **Google Chrome, Apple Safari,** and **UC Browser**.

The Firefox version 2.0 was released in October 2006. This latest version came with new features, such as:

- It has a mail component that is called Thunderbird.
- It provides a quick link to open the Google search engine.
- It has the ability to search multiple search engines simultaneously.
- It provides an efficient user interface.
- It has improved tabbed browsing.
- It offers new security features, including anti-phishing protection.

Google Chrome

Google Chrome is an open-source and the most popular internet browser that is used for accessing the information available on the World Wide Web. It was developed by **Google** on **11 December 2008** for Windows, Linux, Mac OS X, Android, and iOS operating systems. It uses sandboxing-based approach to provide Web security. Furthermore, it also supports web standards like HTML5 and CSS (cascading style sheet).

Google Chrome was the first web browser that has a feature to combine the search box and address bar, that was adopted by most competitors. In 2010, Google introduced the **Chrome Web Store**, where users can buy and install Web-based applications.

What is a search engine?

Search engines allow users to search the internet for content using keywords. Although the market is dominated by a few, there are many search engines that people can use. When a user enters a query into a search engine, a search engine results page (SERP) is returned, ranking the found pages in order of their relevance. How this ranking is done differs across search engines.

Search engines often change their algorithms (the programs that rank the results) to improve user experience. They aim to understand how users search and give them the best answer to their query. This means giving priority to the **highest quality** and most **relevant** pages.

How do search engines work?

There are three key steps to how most search engines work:

- **Crawling** - search engines use programs, called spiders, bots or crawlers, to scour the internet. They may do this every few days, so it is possible for content to be out-of-date until they crawl your website again.
- **Indexing** - the search engine will try to understand and categorise the content on a web page through 'keywords'. Following SEO best practice will help the search engine understand your content so you can rank for the right search queries.
- **Ranking** - search results are ranked based on a number of factors. These may include keyword density, speed and links. The search engine's aim is to provide the user with the most **relevant** result.

Although most search engines will provide tips on how to improve your page ranking, the exact algorithms used are well guarded and change frequently to avoid misuse. But by following search engine optimisation (SEO) best practice you can ensure that:

- Search engines can easily crawl your website. You can also prompt them to crawl new content.
- Your content is indexed for the right keywords so it can appear for relevant searches.
- Your content can rank highly on the SERP.

Directory search engines

Some niche search engines operate as directories for specific types of content. This mean that they only show results for content that is manually added. They do not crawl the internet. SEO tactics can still be used to rank highly for relevant queries within these directory search engines. See types of search engine.

The three of the most popular search engines are:

- **Google** - Google assesses the value of a webpage based on the number of backlinks ie links back to your website. Links from pages that are seen as important by Google weigh more heavily and increase the ranking of the linked pages. Google also analyses the relevancy of the content on the page and other factors like mobile-friendliness. The Google ranking algorithm changes constantly. It can be useful to keep up to date with the latest changes.
- **Bing** - Microsoft's Bing search engine ranks websites based on the webpage content, the number and quality of websites that link to your pages, and the relevance of your website's content to keywords.
- **Yahoo!** - Yahoo! is now powered by Bing.

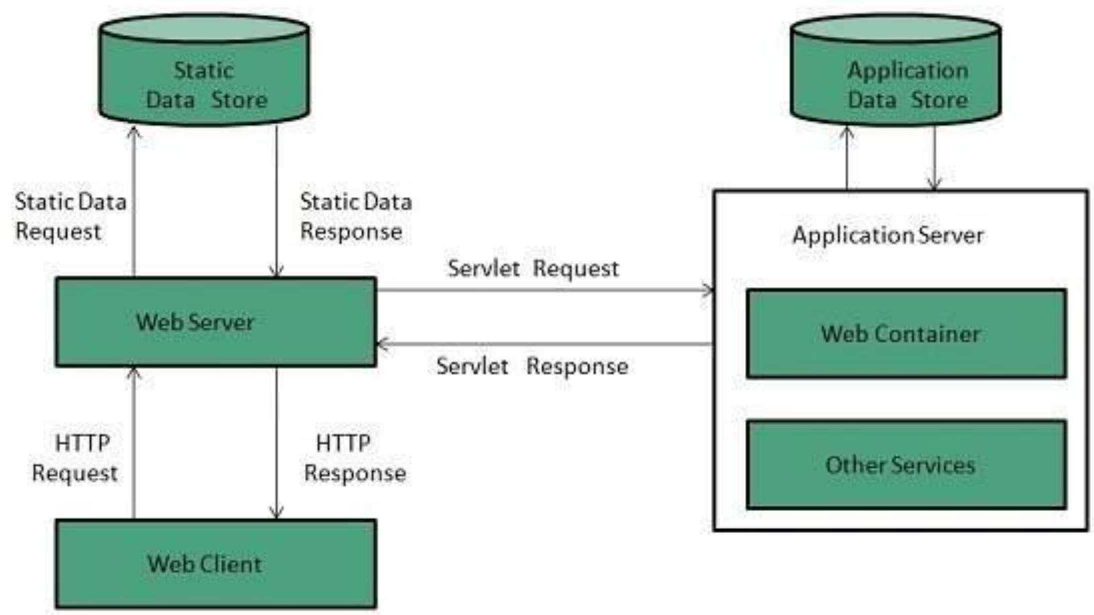
Web server is a computer where the web content is stored. Basically web server is used to host the web sites but there exists other web servers also such as gaming, storage, FTP, email etc.

Web site is collection of web pages whileweb server is a software that respond to the request for web resources.

Web Server Working

Web server respond to the client request in either of the following two ways:

- Sending the file to the client associated with the requested URL.
- Generating response by invoking a script and communicating with database



Key Points

- When client sends request for a web page, the web server search for the requested page if requested page is found then it will send it to client with an HTTP response.
- If the requested web page is not found, web server will the send an **HTTP response:Error 404 Not found.**
- If client has requested for some other resources then the web server will contact to the application server and data store to construct the HTTP response.

Architecture

Web Server Architecture follows the following two approaches:

1. Concurrent Approach
2. Single-Process-Event-Driven Approach.

Concurrent Approach

Concurrent approach allows the web server to handle multiple client requests at the same time. It can be achieved by following methods:

- Multi-process
- Multi-threaded
- Hybrid method.

Multi-processing

In this a single process (parent process) initiates several single-threaded child processes and distribute incoming requests to these child processes. Each of the child processes are responsible for handling single request.

It is the responsibility of parent process to monitor the load and decide if processes should be killed or forked.

Multi-threaded

Unlike Multi-process, it creates multiple single-threaded process.

Hybrid

It is combination of above two approaches. In this approach multiple process are created and each process initiates multiple threads. Each of the threads handles one connection. Using multiple threads in single process results in less load on system resources.

Examples

Following table describes the most leading web servers available today:

S.N.	Web Server Descriptino
1	**Apache HTTP Server** This is the most popular web server in the world developed by the Apache Software Foundation. Apache web server is an open source software and can be installed on almost all operating systems including Linux, UNIX, Windows, FreeBSD, Mac OS X and more. About 60% of the web server machines run the Apache Web Server.

2.	**Internet Information Services (IIS)** The Internet Information Server (IIS) is a high performance Web Server from Microsoft. This web server runs on Windows NT/2000 and 2003 platforms (and may be on upcoming new Windows version also). IIS comes bundled with Windows NT/2000 and 2003; Because IIS is tightly integrated with the operating system so it is relatively easy to administer it.
3.	**Lighttpd** The lighttpd, pronounced lighty is also a free web server that is distributed with the FreeBSD operating system. This open source web server is fast, secure and consumes much less CPU power. Lighttpd can also run on Windows, Mac OS X, Linux and Solaris operating systems.
4.	**Sun Java System Web Server** This web server from Sun Microsystems is suited for medium and large web sites. Though the server is free it is not open source. It however, runs on Windows, Linux and UNIX platforms. The Sun Java System web server supports various languages, scripts and technologies required for Web 2.0 such as JSP, Java Servlets, PHP, Perl, Python, and Ruby on Rails, ASP and Coldfusion etc.
5.	**Jigsaw Server** Jigsaw (W3C's Server) comes from the World Wide Web Consortium. It is open source and free and can run on various platforms like Linux, UNIX, Windows, and Mac OS X Free BSD etc. Jigsaw has been written in Java and can run CGI scripts and PHP programs.

Proxy server is an intermediary server between client and the internet. Proxy servers offers the following basic functionalities:

- Firewall and network data filtering.
- Network connection sharing
- Data caching

Proxy servers allow to hide, conceal and make your network id anonymous by hiding your IP address.

Purpose of Proxy Servers

Following are the reasons to use proxy servers:

- Monitoring and Filtering
- Improving performance
- Translation

- Accessing services anonymously
- Security

Monitoring and Filtering

Proxy servers allow us to do several kind of filtering such as:

- Content Filtering
- Filtering encrypted data
- Bypass filters
- Logging and eavesdropping

Improving performance

It fasten the service by process of retrieving content from the cache which was saved when previous request was made by the client.

Translation

It helps to customize the source site for local users by excluding source content or substituting source content with original local content. In this the traffic from the global users is routed to the source website through Translation proxy.

Accessing services anonymously

In this the destination server receives the request from the anonymzing proxy server and thus does not receive information about the end user.

Security

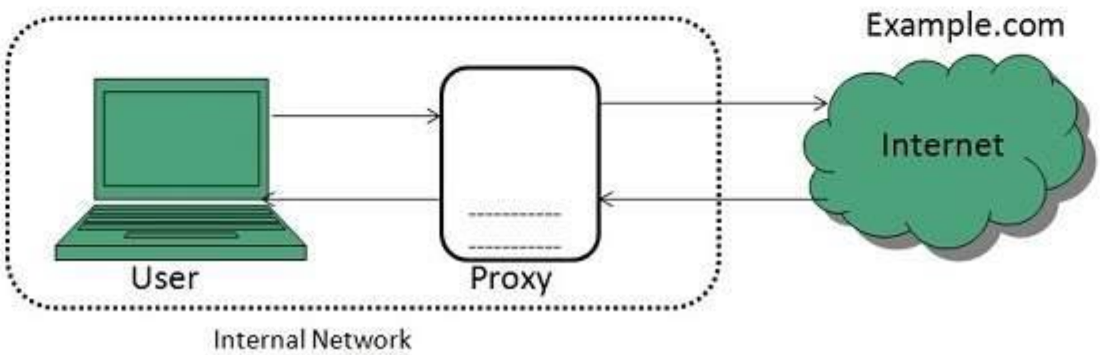
Since the proxy server hides the identity of the user hence it protects from spam and the hacker attacks.

Type of Proxies

Following table briefly describes the type of proxies:

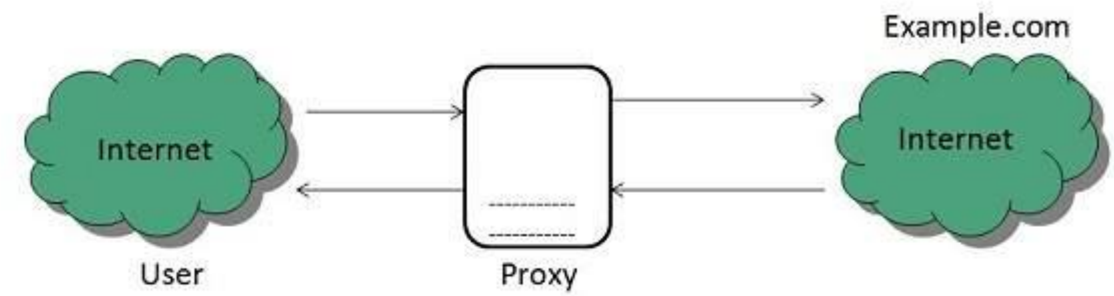
Forward Proxies

In this the client requests its internal network server to forward to the internet.



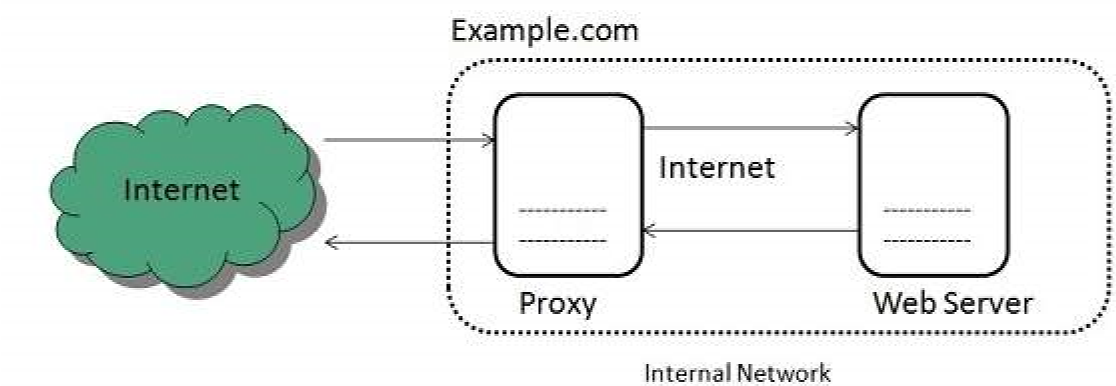
Open Proxies

Open Proxies helps the clients to conceal their IP address while browsing the web.



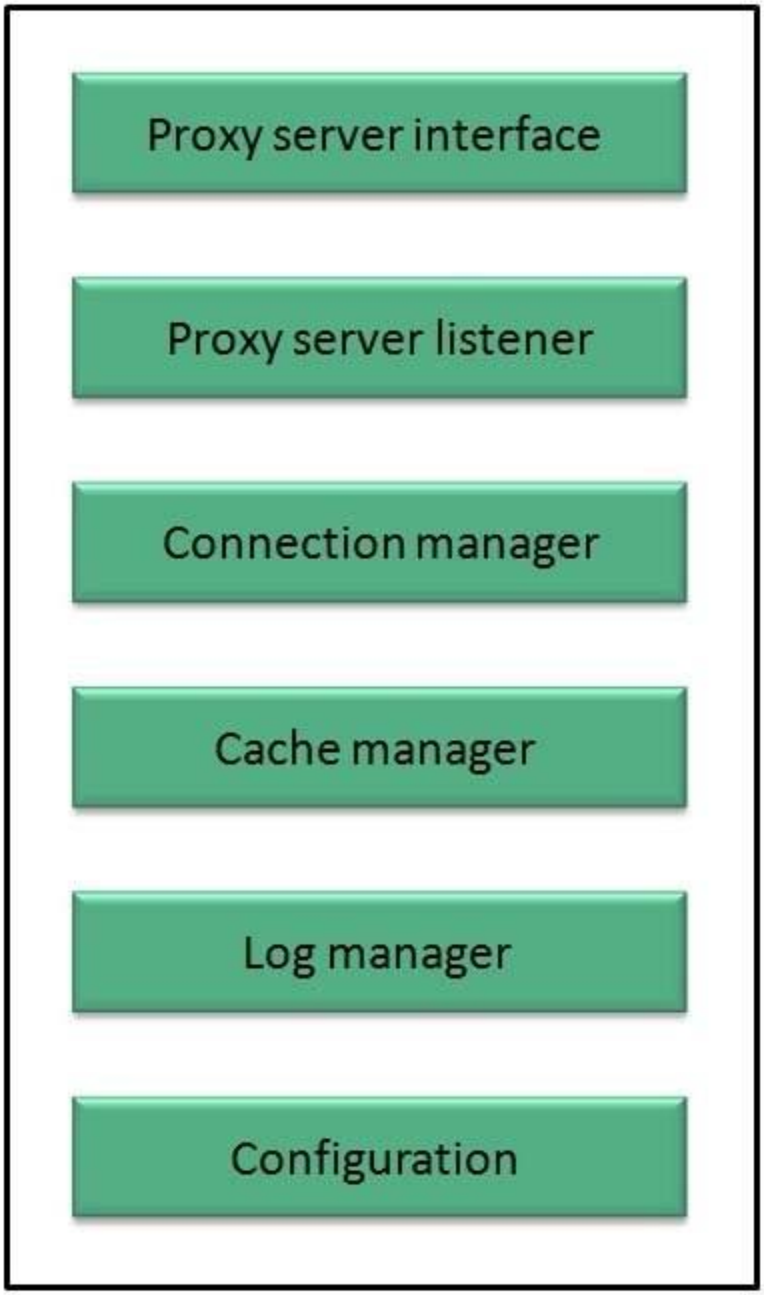
Reverse Proxies

In this the requests are forwarded to one or more proxy servers and the response from the proxy server is retrieved as if it came directly from the original Server.



Architecture

The proxy server architecture is divided into several modules as shown in the following diagram:



Proxy user interface

This module controls and manages the user interface and provides an easy to use graphical interface, window and a menu to the end user. This menu offers the following functionalities:

- Start proxy
- Stop proxy
- Exit
- Blocking URL

- Blocking client
- Manage log
- Manage cache
- Modify configuration

Proxy server listener

It is the port where new request from the client browser is listened. This module also performs blocking of clients from the list given by the user.

Connection Manager

It contains the main functionality of the proxy server. It performs the following functions:

- It contains the main functionality of the proxy server. It performs the following functions:
- Read request from header of the client.
- Parse the URL and determine whether the URL is blocked or not.
- Generate connection to the web server.
- Read the reply from the web server.
- If no copy of page is found in the cache then download the page from web server else will check its last modified date from the reply header and accordingly will read from the cache or server from the web.
- Then it will also check whether caching is allowed or not and accordingly will cache the page.

Cache Manager

This module is responsible for storing, deleting, clearing and searching of web pages in the cache.

Log Manager

This module is responsible for viewing, clearing and updating the logs.

Configuration

This module helps to create configuration settings which in turn let other modules to perform desired configurations such as caching.

What is Apache?

Apache, an open-source Web server created by American software developer Robert McCool. Apache was released in 1995. In the early 2020s, Apache servers deployed about 30 percent of the Internet's content, second only to Nginx.

As a Web server, Apache is responsible for accepting directory (HTTP) requests from Internet users and sending them their desired information in the form of files and Web pages. Much of the Web's software and code is designed to work along with Apache's features. Programmers working on Web applications typically make use of a home version of Apache to preview and test code. Apache also has a safe and secure file-sharing feature, allowing users to put files into the root directory of their Apache software and share them with other users. The Apache server's impact on the open-source software community is partly explained by the unique license through which software from the Apache Software Foundation is distributed.

Apache was originally known as the NCSA HTTPd Web server and was written by McCool when he was an undergraduate at the National Center for Supercomputing Applications (NCSA) at the University of Illinois at Urbana-Champaign. Apache is maintained and developed by a large community of volunteers and developers from the Apache Software Foundation, as well as by contributions from users worldwide

What is IIS?

The term "IIS" stands for Internet Information Services, which is a general-purpose webserver that runs on the Windows operating system. The IIS accepts and responds to the client's computer requests and enables them to share and deliver information across the LAN (or Local Area Network) such as a corporate intranet and the WAN (or Wide Area Network) the internet. It hosts the application, websites, and other standard services needed by users and allows developers to make websites, applications and virtual directories to share with their users. A web server provides the users with information in several different forms, such as File exchanges as a download, uploads, Images files, HTML pages, and text documents. The webservers are commonly used as a portal for sophisticated and highly interactive websites, applications that tie middleware and back-end applications together to make enterprise-grade-systems. For example, AWS enables media services such as Netflix to provide real-time streaming content. Amazon web services also enable public cloud administration all through the webservers. Generally, the IIS is also compared with the Apache, which is also a kind of web server that is freely available for everyone. We can simply say that both works the same except that the apache web server can be used almost on any operating system such as Windows, Linux, and Mac, While the IIS is only available for windows. However, the IIS integrates with Microsoft's other products, such as the .NET Framework, the ASP scripting language. The IIS also has its own helpdesk to manage and solve issues while, on the other hand, the Apache webserver's supports almost come from the user community. Additionally, the IIS has the security features, which makes it a more secure and efficient option than the Apache.

How IIS works

It works through several different standard languages and protocols. HTML is used for creating a variety of elements. For example, texts, buttons, hyperlinks, and direct/indirect behaviors. The HTTP (or Hyper Text Transfer Protocol) is used for exchanging the information between the two or more servers and users. HTTPS --HyperText Transfer Protocol Secure over the SSL (or Secure Sockets Layer) -- uses SSL (secure sockets layer) to encrypt the communication to add additional data security. The FTP (or File Transfer Protocol), or its secure variant, FTPS, can transfer files.

IIS Express for testing

Microsoft itself provides a specific IIS version, which is known as the IIS Express, for developers to test their websites and applications. IIS Express provides all the major capabilities of a full IIS web server, but many tasks can be performed without administrative privileges.

Security

Organizations need to take security measures to protect the webserver from security breaches to ensure that the website is secure. Companies can use the facilities built into IIS to harden IIS.

Some of the ways that can be used to harden the IIS to avoid the security breaches are listed below:

1. Configuration of error pages should be done in such a way that they will display only relevant information about the issues received. The error pages do not display unnecessary information such as IP addresses of servers, user IDs and passwords or any other type of information that can help hackers in exploiting the webserver.
2. The "URL authorization" must be used in order to apply rules for specific requests e.g., dealing with a particular kind of URLs. URL authorization allows a company to authorize only certain users to view the requested pages.
3. Any feature of IIS that does not help in reducing the potential attack should be disabled.
4. The access of domains and IP addresses must be controlled that can reach the webserver.

5. Always use the firewall to ensure that only valid data package can reach the server.
6. Whenever Windows gets an update, the Windows operating system should be updated with the latest security patches.
7. The logging must be used to manage the record of the visitors that access the webserver.